

Advances in reinforcement learning for traffic signal control: a review of recent progress

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In recent years, reinforcement learning (RL) methods have shown substantial promise in traffic signal control (TSC), drawing significant attention from academia and industry. The integration of RL with advanced machine learning methods offers a novel approach to tackling dynamic and complex traffic conditions. Previous reviews have primarily focused on foundational RL methods or controlled experimental setups, with limited attention to advanced methodologies and real-world complexities. To address this gap, this paper provides a comprehensive review of recent advancements in RL-based TSC from both methodological and application perspectives. From the methodological perspective, it comprehensively examines how advanced RL methods are integrated with state-of-the-art artificial intelligence algorithms to address TSC challenges. From the application perspective, it systematically reviews studies that extend RL approaches to more complex and multi-modal traffic scenarios. Furthermore, it identifies critical challenges, including model scalability, real-time adaptability, the sim-to-real gap, and the integration of TSC with broader urban transport systems, offering insights, and directions for future research.

Keywords: reinforcement learning; traffic signal control; artificial intelligence; literature review.

1. Introduction

1.1. Background

With the rapid development of society and the substantial growth in the number of vehicles, traffic congestion has become a global challenge. Traffic signal control (TSC) systems are crucial for mitigating congestion and enhancing the efficiency and safety of urban traffic networks. Traditional TSC methods can be broadly classified into three categories: fixed-time control, actuated control, and adaptive control. Fixed-time control uses predetermined timing schedules, offering simplicity but lacking flexibility in dynamic traffic conditions. Actuated control adjusts signal phases based on real-time detection, providing more adaptability but facing challenges in handling sustained congestion and fluctuating traffic patterns, due to its reliance on short-term data. Adaptive control is the most advanced technology, interacting dynamically with traffic conditions to continuously adjust signal timing strategies in response to complex, unfamiliar scenarios. However, adaptive control systems still face several practical limitations. First, these systems typically rely on predefined optimization models, which are insufficient for responding to random traffic demands and unexpected incidents. Second, when managing large-scale traffic networks, especially in highly congested or oversaturated conditions, the optimization methods of current systems are insufficient to effectively cope with the challenges. Overall, these methods are effective in simpler scenarios but encounter significant limitations in managing complex urban traffic environments such as high traffic volumes,

multimodal traffic, and coordination across intersections. These issues significantly limit the performance of current systems in addressing real-world traffic situations, and there arises an urgent demand for more intelligent and adaptive solutions.

1.2. Opportunities

With the rapid advancement of artificial intelligence (AI) technologies, reinforcement learning (RL) has emerged as a powerful tool for addressing TSC challenges. Compared to traditional methods, RL offers several notable advantages. Firstly, its capacity for self-learning and adaptability to dynamic environments is particularly notable. RL autonomously learns optimal control strategies through environmental interaction, without relying on predefined models or rules. By continuously adapting based on real-time traffic data, RL adjusts to evolving traffic conditions, making it well-suited for complex and dynamic traffic scenarios. Secondly, RL's sequential decision-making framework is particularly suited for traffic control, where decisions at one intersection can affect others over time. By optimizing for long-term impacts across multiple intersections, RL ensures globally coordinated decision-making, avoiding local optima and improving overall network efficiency. Lastly, RL's scalability is robust, enabling integration with deep learning models to address more complex problems. The introduction of deep reinforcement learning (DRL) has further elevated RL's applications in TSC by combining RL's decision-making with the powerful representational capacity of deep neural networks. This integration allows RL to operate in high-dimensional state

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spaces and manage more intricate traffic scenarios. Moreover, RL presents new opportunities for controlling multimodal traffic flows. While traditional methods can manage multimodal transportation to some extent, they often encounter limitations when addressing the dynamic and complex demands of various traffic modes simultaneously. RL, with its unified and adaptive framework, offers enhanced capabilities to optimize the control of motor vehicles, pedestrians, cyclists, and public transport, significantly improving the efficiency of the entire traffic network.

1.3. Motivation and scope

With the acceleration of urbanization and growing transportation demand, RL-based TSC has become increasingly important in traffic engineering and intelligent transportation systems. RL methods not only effectively address the complex and dynamic traffic conditions that existing traffic light control systems struggle to manage, but also offer promising solutions for practical engineering applications. Recent studies have demonstrated the considerable research value of utilizing RL to enhance TSC. However, this field is highly interdisciplinary, blending AI with traffic engineering, thus presenting a technological barrier for scholars and practitioners. Therefore, a systematic review and analysis of this field is needed to guide both practical traffic engineering and academic research.

A number of review articles have explored RL applications in TSC. Haydari and Yimaz (2022) provided a detailed overview of early RL research in traffic control. Li (2023a) systematically analyzed RL applications across various traffic scenarios. Wei (2021) comprehensively summarized RL-based TSC studies up until 2020. Noaeen (2022) focused on network-level RL applications in TSC, emphasizing statistical and methodological analyses. Miletić (2022) reviewed RL methodologies from the perspective of single and multi-intersection traffic control. Chen et al. (2022) discussed RL applications considering practical engineering challenges, and Koohy (2022) examined reward function design in RL-based traffic light control. Despite these existing reviews, several significant limitations remain:

- 1) Gap between methodological updates and recent advancements: Existing reviews primarily focus on research prior to 2022, while significant innovations have emerged in TSC between 2022 and 2025. Particularly, large language model (LLM)-based reinforcement learning applications have not been addressed in previous reviews at all.
- 2) Lack of innovative application scenarios: Current reviews predominantly concentrate on simple traffic scenarios, lacking systematic analysis of emerging complex multimodal traffic networks, such as mixed traffic environments incorporating shared mobility, micro-mobility vehicles, and autonomous vehicles.
- 3) Insufficient engineering implementation perspective: Existing literature provides limited analysis from an engineering implementation standpoint regarding the challenges of deploying reinforcement learning in real traffic systems, especially adaptability issues across different urban environments.

This review aims to address the gaps in the current literature by offering a more comprehensive and in-depth perspective. Specifically, the contributions of this paper are reflected in the following unique aspects:

- 1) This review systematically examines advanced RL methods integrated with state-of-the-art AI algorithms, investigating

how these approaches can be optimized to address the critical challenges and unique features of TSC from an engineering perspective.

- 2) The review extends the focus of RL approaches from simple experimental setups to more complex, multimodal traffic networks, assessing their real-world applications.
- 3) This review pinpoints the primary challenges in current literature, including model scalability, real-time adaptability, the sim-to-real gap, and the integration of TSC with other transport systems for coordinated optimization.

Through these discussions, we aim to provide novel insights and practical guidance for scholars and practitioners in traffic engineering and AI, fostering the development and application of RL-based TSC in complex real-world scenarios. The rest of the paper is organized as follows: Section 2 describes the literature review methods, including sources, keywords, and analysis of publication trends. Section 3 covers the fundamentals of reinforcement learning in TSC, introducing basic concepts and key algorithms. Section 4 discusses recent advances in RL methods, focusing on emerging AI methods. Section 5 explores RL applications in complex traffic scenarios and practical challenges encountered in real-world deployments. Section 6 addresses challenges and future directions. Section 7 lists available benchmarks, and Section 8 concludes this paper. The specific abbreviations of related terms can be found in Appendix A.

2. Literature review methods

To conduct this literature review, we employed specific search criteria to comprehensively explore the application of advanced reinforcement learning methods in TSC. The keywords "traffic signal" or "traffic light" and "reinforcement learning" were used to filter papers based on their titles and abstracts. The primary databases consulted were Scopus and Web of Science, known for their broad coverage of high-impact journals and conferences. Additionally, we utilized Google Scholar, IEEE Xplore, ACM Digital Library, and Springer Link to capture more specialized publications, particularly in intelligent systems and computational methodologies. The search was conducted between January 2015 and March 2025, with 2015 chosen as the starting point because it marks a significant shift in reinforcement learning research. Specifically, the release of Deep Q-network (DQN) in 2015 (Mnih, 2015) was a pivotal development that spurred significant progress in applying reinforcement learning to control systems, including TSC. While earlier studies on the topic do exist, our review focuses on the exploration of advanced reinforcement learning methods, and as such, we deliberately chose to begin the search from 2015, when these more sophisticated techniques began to emerge. After manually filtering out studies unrelated to TSC, a total of 470 papers were selected for detailed analysis.

Figure 1 illustrates the publication trends over the years, reflecting the continuous interest in TSC research within the reinforcement learning community. The number of publications has grown steadily, highlighting the increasing focus on this area. The slight dip in 2025 publications may be due to delays in indexing, particularly for papers from international conferences that have yet to be fully captured in the databases. This explains the apparent decline in 2025, even though research in this field remains active.

Figure 2 showcases the top 10 journals with the highest number of published articles. The *IEEE Transactions on Intelligent Transportation Systems* stands out as the journal with the highest

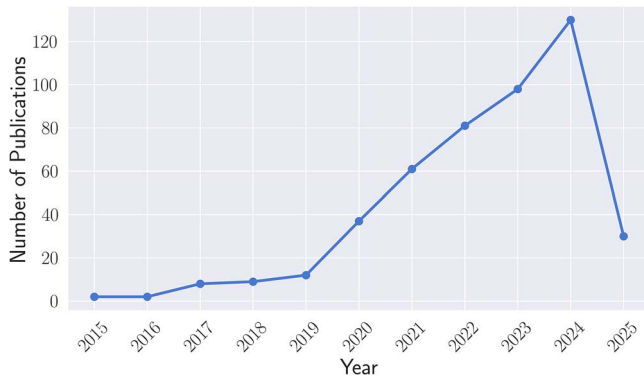


Figure 1. Publications trend per year.

number of publications, reflecting the close relationship between TSC and intelligent transportation systems. Additionally, journals such as *Transportation Research Part C: Emerging Technologies*, and *IEEE Transactions on Vehicular Technology* demonstrate the strong presence of research that bridges both transportation engineering and the development of emerging technologies. Interestingly, several of the top publishing venues include broader AI and engineering journals, such as *IEEE Access*, *Expert Systems with Applications* and *Engineering Applications of Artificial Intelligence*. This indicates that research on reinforcement learning for TSC is not confined to the transportation domain but is also seen as a key topic within the broader field of AI and its applications in engineering systems. The variety of journals publishing in this area suggests that the field is interdisciplinary, requiring expertise not only in transportation systems but also in the development and application of sophisticated reinforcement learning algorithms.

3. Fundamentals of reinforcement learning in traffic signal control

Section 3 introduces the concepts of TSC, traditional control methods, as well as the fundamental principles and key algorithms for formulating TSC as a RL problem.

3.1. Background of traffic signal control

3.1.1. Basic concepts

TSC systems primarily aim to efficiently manage vehicular and pedestrian flow at intersections. Key concepts include intersections, traffic signals, phases, and lanes, which collectively form the foundation of TSC operations. A schematic diagram of an intersection is shown in Fig. 3.

3.1.2. Intersection

An intersection is where two or more roads meet, allowing vehicles and pedestrians to cross or turn. Intersections can operate independently (isolated intersections) or be part of a coordinated system with multiple intersections (network-level) to optimize overall traffic flow.

3.1.3. Traffic signals

Traffic signals are crucial devices for managing traffic flow, using displays of red, yellow, and green lights to indicate when to proceed, slow down, or stop. A single traffic signal often consists of multiple phases, with each phase representing a combination of light displays tailored to varying traffic conditions, thereby controlling the movement of vehicles and pedestrians.

Phase: A phase is a specific combination of traffic signal displays that determines which lanes are permitted to proceed and for how long. Typically, a phase corresponds to the green light for straight-through, left-turn, or right-turn movements in a particular direction.

Lane: A lane is a designated path on a road for vehicle movement, often categorized by direction and function, such as straight, left-turn, and right-turn lanes. Each lane is typically associated with a specific phase, ensuring that vehicles can proceed smoothly according to the traffic signal indications.

3.1.4. Traditional traffic control methods

Fixed-time signal control (FTSC) sets predetermined signal cycles and phase durations, adjusting schedules based on peak and off-peak hours. It is simple and cost-effective but struggles with real-time adaptability, making it less suitable for unexpected traffic changes.

Traffic actuated signal control (TASC) uses detectors to gather real-time traffic data, adjusting phases dynamically based on vehicle arrivals. While more responsive than FTSC, TASC faces coordination challenges in dense networks, often resulting in inefficient stop-and-go patterns between adjacent intersections.

Adaptive traffic signal control (ATSC) improves upon TASC by incorporating real-time traffic predictions and adjusting signals accordingly. Systems like SCOOT (Robertson and Bretherton, 1991) and SCATS (Sims and Dobinson, 1980) optimize traffic flow across multiple intersections, but their reliance on preset rules limits flexibility in complex and dynamic traffic situations.

Each traditional method has its limits: FTSC lacks real-time flexibility, TASC struggles with multi-intersection coordination, and ATSC's adaptability is constrained by predefined rules. In contrast, RL offers self-learning capabilities, allowing dynamic adjustments to traffic conditions without extensive preset rules, making it more effective for managing complex and variable scenarios.

3.1.5. Evaluation metrics

In the research and practical application of TSC, evaluation metrics are employed to assess the performance and optimization efficacy of control systems. Commonly used metrics include:

- **Delay time:** Delay time refers to the waiting period that vehicles experience at an intersection caused by traffic signals. It is a critical indicator of traffic flow efficiency, with shorter delays indicating smoother vehicular passage.
- **Throughput:** Throughput measures the number of vehicles that pass through an intersection within a given period. A higher throughput signifies greater efficiency in managing traffic flow at the intersection.
- **Queue length:** Queue length denotes the length of the vehicle queue waiting at a traffic signal.
- **Waiting time:** Waiting time captures the duration that a vehicle remains stationary or moves at a very low speed due to traffic signals or congestion at the intersection.
- **Travel time:** Travel time represents the total duration a vehicle takes to traverse from entering to exiting the intersection.

A comprehensive summary of studies and implementations employing these metrics, along with their performance evaluations across various methods, is provided by Mei (2024). In practice, to obtain these metrics, intersections typically utilize various sensors:

Traditional sensors: These include inductive loops, infrared sensors, microwave radars, and ultrasonic sensors. They detect

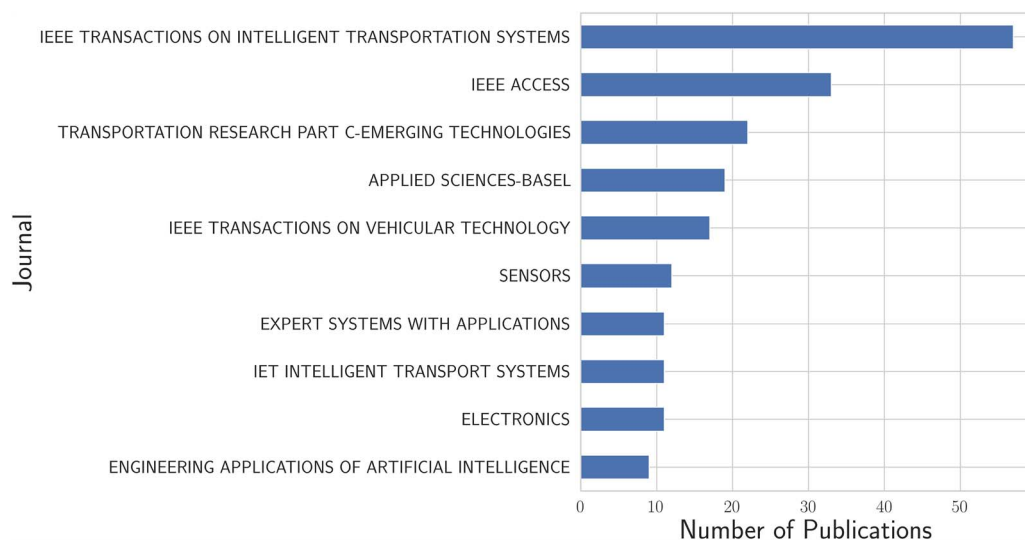


Figure 2. Top 10 journals by number of publications.

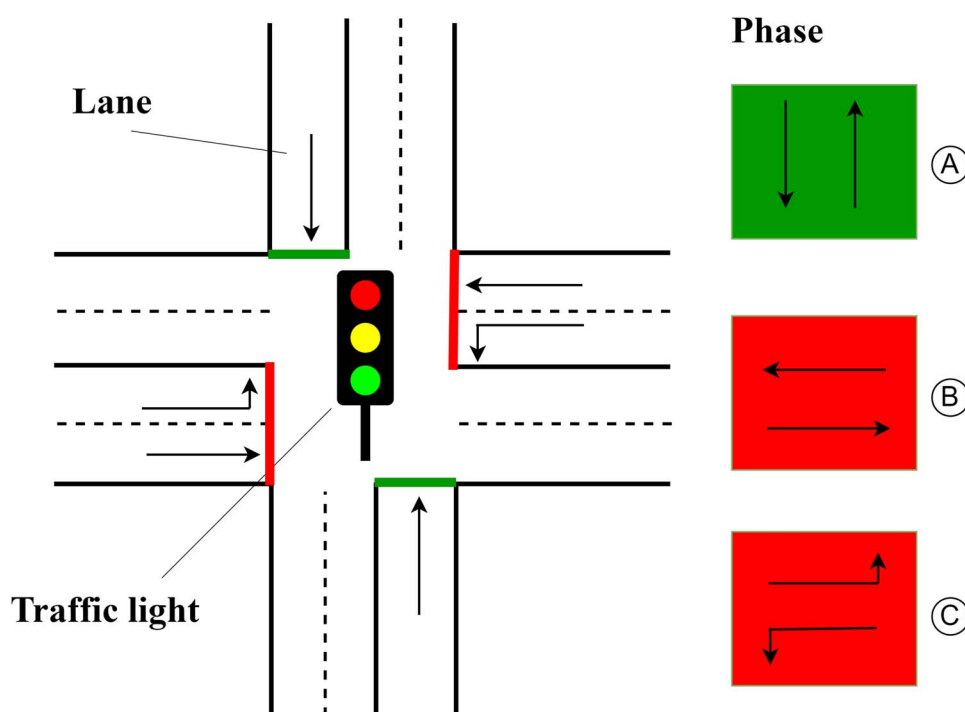


Figure 3. Illustration of key elements in a signalized intersection.

vehicle presence, speed, and volume, providing real-time data for signal adjustments. However, their installation can be costly due to the need for road surface modifications.

Emerging computer vision technologies (Shirazi et al., 2023): Video detection systems using cameras and algorithms offer a versatile alternative, capturing detailed data such as queue lengths and vehicle trajectories. These systems provide lower installation costs and support real-time analysis, making them suitable for adaptive signal control in cloud or edge computing environments. However, they also have certain limitations, including susceptibility to environmental lighting conditions (e.g. reduced accuracy in rain or low-light scenarios), high computational resource requirements for data processing, and potential concerns regarding privacy protection.

3.1.6. Traffic simulation environment

Traffic simulation uses computer models to emulate real-world traffic flow, aiming to analyze vehicle behavior and traffic flow characteristics under different conditions by virtually replicating vehicles, roads, and traffic signals. In TSC, microscopic traffic simulation is particularly crucial as it models the behavior and interactions of individual vehicles. These simulations provide a precise evaluation platform for designing and optimizing signal control strategies, making them particularly important for reinforcement learning-based TSC. Currently, the most widely used microscopic traffic simulation tools are SUMO (Lopez, 2018) and CityFlow (Zhang, 2019), both of which are open-source. Recently, CityFlowER (Da 2024a) was introduced as an extension of CityFlow, designed for enhanced efficiency and realism by

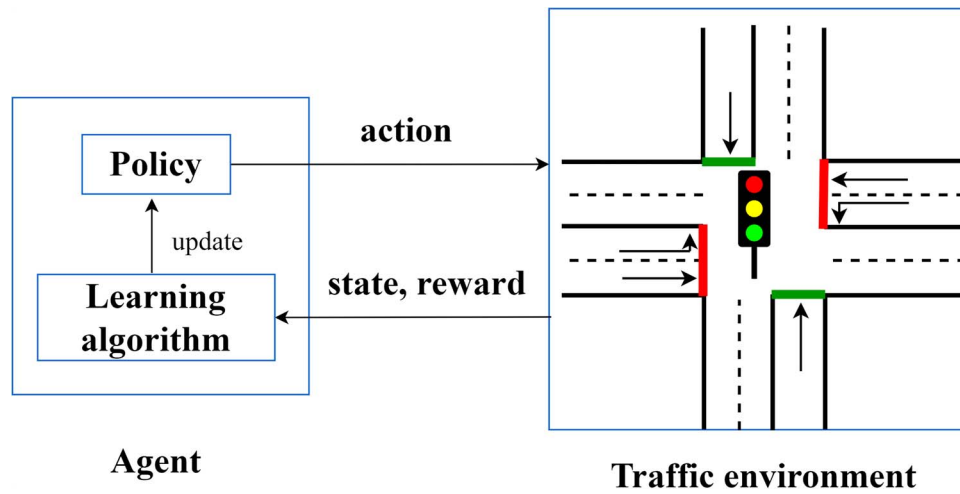


Figure 4. Illustration of the learning interaction between the RL agent and traffic environment.

embedding machine learning models directly into the simulation core, potentially accelerating research involving large numbers of ML-controlled agents. Apart from SUMO and CityFlow, other popular microscopic traffic simulation software includes VISSIM and AIMSUN. However, as these are generally commercial software, their restricted usage limits accessibility for open-source research and reinforcement learning training compared to SUMO and CityFlow.

3.2. Traffic signal control as a reinforcement learning problem

3.2.1. Markov decision process

The interaction between the reinforcement learning agent and the traffic environment is crucial for the learning process, as illustrated in Fig. 4. The core of formulating the TSC problem as a RL problem lies in defining the Markov decision process (MDP), which is composed of a quintuple (S, A, P, R, γ) :

State (S)

The state S captures the current traffic conditions at an intersection. A review by Rasheed et al. summarizes various state definitions, with the most common being the number of vehicles on lanes. Other definitions include real-time image and video data (Jeon et al., 2018; Wei, 2018; An and Zhang, 2022; Chu et al., 2022; Ma, 2022), position and velocity grids (Sahu, 2021; Liu et al., 2022), which partition the area around the intersection into grids to capture the spatial distribution and motion characteristics of vehicles, such as their number and speed, efficient phase (Zhang 2022b), vehicle dynamics like speed and density (Kumar, 2022), and GPS trajectories of vehicles (Mohamad Alizadeh and Abdulhai, 2022).

Reward (R)

The reward R measures the system's performance after executing an action and guides the optimization of traffic signals. As summarized by Koohy (2022), some commonly used reward functions include:

- Average speed: Aims to maximize vehicle throughput by reducing idling or slow-moving times, and is calculated based on the average speed of vehicles.
- Average speed normalized: Normalizes the average speed by considering the maximum speed of different road segments to account for speed limit variations.
- Maximum wait time: Prioritizes minimizing the delay of the longest waiting vehicle.

- Aggregate wait time: Uses the negative value of the total waiting time of all queued vehicles to minimize overall delays.
- Aggregate wait time normalized: Similar to aggregate wait time but normalized to avoid a strict first-in-first-out queuing policy.
- Pressure: Defined as the difference in vehicle density between upstream and downstream lanes, it balances traffic flow to reduce congestion.
- Queue length: Uses the number of queued vehicles as the reward, aiming to minimize queue length.
- Maximum wait aggregated queue: Combines the maximum wait time and queue length to approximate the worst cumulative waiting time for all vehicles.

The primary distinction between evaluation metrics and reward functions lies in their temporal focus: evaluation metrics are cumulative and assess a system's overall performance over a period of time, whereas reward functions operate step-wise, providing real-time feedback to guide optimization during training. Notably, certain metrics, such as queue length and waiting time, are occasionally adapted as reward functions in reinforcement learning frameworks, often with modifications like normalization or aggregation. While reinforcement learning frameworks are fundamentally capable of handling delayed rewards, in the context of TSC, directly using metrics like travel time as the reward signal during training is less common. This is often because estimating or obtaining accurate travel time feedback step-by-step can be complex and may introduce significant delays or noise into the learning process. Therefore, researchers frequently prefer reward functions based on more immediately available state information, such as queue length or waiting time, to provide denser feedback for agent training. However, metrics like travel time remain crucial and are widely used as evaluation metrics to assess the overall system performance and effectiveness of the final learned policy. The reward function can be expressed as $r_t = R(s_t, a_t)$, or in a multi-intersection setting, $r_t^i = R_i(s_t^i, a_t^i)$, where r_t^i represents the reward at intersection i at time t .

Action (A)

The action A pertains to the signal control operations that the traffic system can execute, typically including:

- 1) Cyclic: Signal phases transition sequentially in a predefined fixed cycle, with actions represented as continuous values.

Actions determine the green light duration for each phase (Zeng, 2022; Han 2024a).

- 2) Acyclic phase selection: Signal phases do not follow a fixed order. Each intersection has a predefined set of signal phases, and actions are discrete values representing the selection of the most appropriate phase based on current traffic conditions. Phase selection occurs at fixed decision intervals, with the chosen phase remaining unchanged until the start of the next interval (Celtek et al., 2021).
- 3) Hybrid actions: This action space combines discrete phase selection with continuous duration optimization. It allows simultaneous selection of the next active phase and adjustment of its green light duration, without relying on fixed cyclic transitions and fixed decision intervals (Luo et al., 2024; Fan, 2025).

State Transition Probability (P)

Defines the probability of transitioning from the current state s_t to the next state s_{t+1} after taking action a_t , denoted as $P(s_{t+1} | s_t, a_t)$. In TSC, state transitions are often deterministic but can also be stochastic due to fluctuations in vehicle dynamics and traffic flow.

Discount factor (γ)

Balances the importance of future rewards against immediate ones, with γ usually in the range $0 \leq \gamma < 1$. A value close to 1 places more emphasis on long-term gains, while a smaller γ focuses on short-term benefits.

3.2.2. Multi-agent reinforcement learning (MARL)

While standard Reinforcement Learning often assumes a fully observable MDP as outlined in Section 3.2.1, MARL methods applied to real-world TSC typically operate under conditions of partial observability. This necessitates modeling the problem as a Partially Observable Markov Decision Process (POMDP), which presents fundamental differences and unique challenges compared to the standard MDP formulation, as highlighted by the reviewer. Specifically, a key difference is that in a POMDP, an agent does not perceive the true underlying system state (s) directly; instead, it receives an observation (o) that is only probabilistically related to the state ($P(o|s)$). This contrasts sharply with the MDP assumption where the agent observes the full state s . In the context of MARL-TSC, this partial observability arises naturally because each traffic signal controller (agent) typically only has access to local information from its immediate intersection (e.g. vehicle counts, queue lengths via sensors). Crucially, agents usually lack access to the complete state of the entire traffic network or the precise status (e.g. queue lengths, intended actions) of other, especially distant, intersections. This inherent information incompleteness means each agent functions under partial observability, making the decision-making process fundamentally a POMDP.

Consequently, applying MARL effectively in POMDPs presents several key challenges: (1) State estimation: agents need to infer hidden state components from local observations. (2) Belief state representation: policies must be based on a belief state (a distribution over possible true states) instead of a single state. (3) Coordination under uncertainty: agents must coordinate their actions despite incomplete knowledge of the true state or the intentions of other agents. While methods like recurrent neural networks (RNNs), belief tracking algorithms, and Dec-POMDP-based MARL approaches exist, addressing these challenges remains an active research area in MARL-TSC.

Considering the spatial and temporal dependencies within traffic networks, multi-agent systems must effectively account for agent interactions to develop coherent and optimized signal control strategies. Within this multi-agent setting, two principal architectures are often employed:

- 1) Centralized training and decentralized execution (CTDE): This architecture leverages global information during training, allowing agents to share state and reward data for more effective cooperative strategy learning. However, during execution, each agent makes decisions based solely on its local information, retaining a decentralized nature (Hu and Li, 2024). The primary advantage of CTDE lies in its capacity to use global insights during training for enhanced coordination and potentially better overall network performance, while the decentralized execution maintains scalability suitable for large-scale traffic network control. However, a key disadvantage is its reliance on global information accessibility during the training phase, which can be challenging or resource-intensive, and the centralized training process itself might face scalability limits as the number of agents grows significantly.
- 2) Decentralized training and decentralized execution (DTDE): This architecture emphasizes that each agent relies exclusively on its local information for decision-making throughout both training and execution (Dai 2024a;). Its main advantages are its inherent scalability for both training and execution, operational flexibility, and significantly reduced communication overhead, making it highly suitable for very large networks or scenarios with strict information access limitations. Conversely, a notable disadvantage is that the lack of information sharing can make it difficult for agents to learn complex coordinated behaviors, potentially resulting in suboptimal joint strategies and reduced overall system efficiency compared to CTDE.

In these architectures, the policy $\pi = (\pi^1, \pi^2, \dots, \pi^n)$ represents the joint policy, where each agent i selects action a_i^t according to its policy π^i . In a CTDE framework, centralized training enables the joint policy to better capture cooperative relationships among agents, as shown in Figs 5 and 6. In contrast, the DTDE framework fosters more independent decision-making, suitable for scenarios where full coordination is less critical or feasible. The choice of architecture depends on the network's scale, communication constraints, performance requirements, and the required complexity of interactions among agents.

3.2.3. Key reinforcement learning algorithms

1) Deep Q-network (DQN)

DQN (Mnih, 2015) is a value-based reinforcement learning algorithm that makes decisions by learning a state-action value function (Q-function). The Q-function $Q(s, a)$ represents the expected cumulative reward the agent can obtain after taking action a in state s . DQN approximates the Q-function using neural networks and updates the network parameters by minimizing the temporal-difference (TD) error:

$$L(\theta) = \mathbb{E}_{(s,a,r,s') \sim D} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2 \right] \quad (1)$$

where θ denotes the parameters of the Q-network, θ^- represents the parameters of the target Q-network, D is the experience replay buffer, γ is the discount factor, and r is the immediate reward.

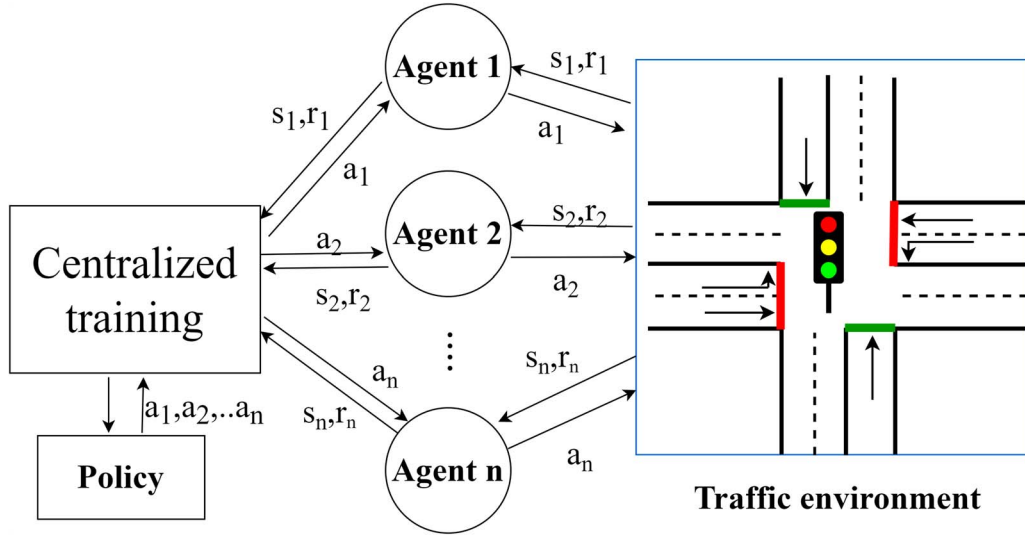


Figure 5. CTDE architecture.

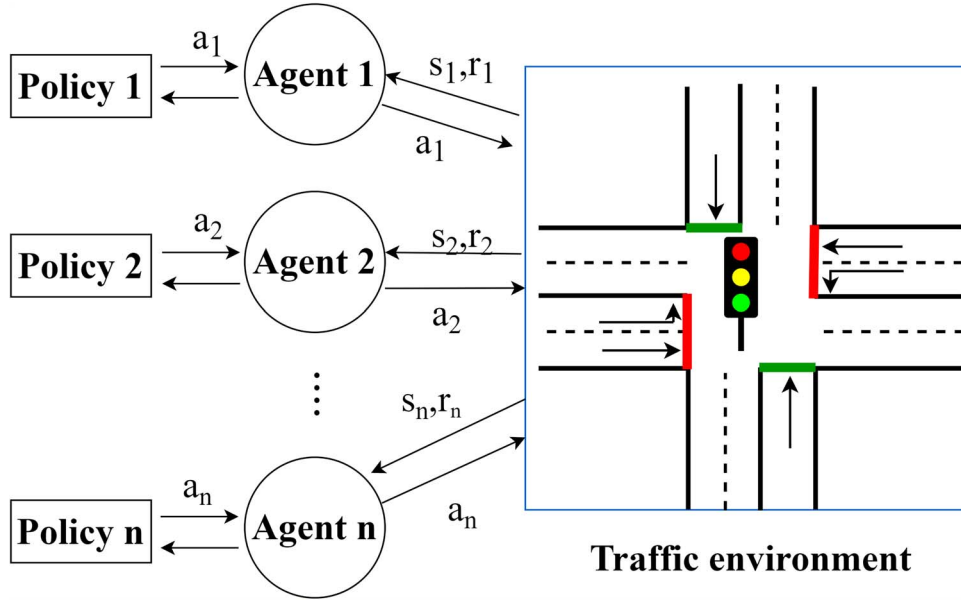


Figure 6. DTDE architecture.

In RL-TSC, DQN optimizes signal phases at intersections by selecting actions that correspond to the maximum Q-value, making it applicable to both single-agent and multi-agent scenarios.

2) Proximal policy optimization (PPO)

PPO (Schulman, 2017) is a policy-based reinforcement learning algorithm that directly optimizes the policy $\pi(a | s; \theta)$ to maximize the cumulative expected reward. PPO maintains stability in policy updates by employing the following objective function to constrain the extent of policy changes:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (2)$$

where $r_t(\theta) = \frac{\pi(a_t | s_t; \theta)}{\pi(a_t | s_t; \theta_{\text{old}})}$ is the ratio between the new and old policies, $\hat{A}_t$ is the advantage function, and ϵ is a clipping hyperparameter.

PPO is particularly effective in RL-TSC for managing both continuous and discrete signal phase control problems, offering stable policy updates and favorable convergence properties.

3) Soft actor-critic (SAC)

SAC (Haarnoja, 2018) is a policy-based algorithm that maximizes the expected reward while encouraging exploration through policy entropy. Its objective function is given by:

$$J(\pi) = \sum_t \mathbb{E}_{(s_t, a_t) \sim \rho_\pi} [r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot | s_t))] \quad (3)$$

where $\mathcal{H}(\pi(\cdot | s_t))$ denotes the entropy of the policy, and α is the coefficient balancing reward and entropy.

By simultaneously learning the policy and the Q-function (critic) and incorporating randomness into the policy updates, SAC is well-suited for RL-TSC scenarios requiring robust exploration capabilities and stable performance in complex environments.

4) Multi-agent proximal policy optimization (MAPPO)

MAPPO (Yu, 2022) extends PPO to multi-agent environments by maintaining independent policies $\pi^i(a_i^t | s_i^t)$ for each agent i . MAPPO facilitates collaborative learning by sharing experiences among agents and adopts the CTDE framework to leverage global

state information during training, enhancing multi-agent strategy optimization.

In RL-TSC, MAPPO is suitable for coordinated control across multiple intersections, enabling agents to learn cooperative strategies to improve overall traffic flow efficiency.

5) Q-mixing network (QMIX)

QMIX (Rashid, 2020) is a multi-agent value decomposition algorithm that represents the joint Q-function $Q_{\text{total}}(s, \mathbf{a})$ as a combinatorial function of local Q-functions for each agent:

$$Q_{\text{total}}(s, \mathbf{a}) = f_{\text{mix}}(Q^1(s^1, a^1), Q^2(s^2, a^2), \dots, Q^n(s^n, a^n)) \quad (4)$$

where f_{mix} is a mixing network that combines local Q-values into the joint Q-value. QMIX leverages the CTDE framework, allowing each agent's policy updates to depend solely on local information, thereby improving training efficiency and distributed decision-making.

In multi-intersection RL-TSC scenarios, QMIX efficiently learns joint policies by decoupling each intersection's Q-function into local components, providing strong collaboration performance in multi-agent environments.

These algorithms offer diverse solutions for reinforcement learning in TSC across both single-agent and multi-agent settings. However, the multi-agent TSC problem presents complexities such as highly coupled network structures and the uncertainty of vehicular flows. While some existing multi-agent algorithms (e.g. MASAC (Mao, 2023), MADQN (Haddad et al., 2022)) show strong performance in RL-TSC, they may not fully capture the unique challenges of TSC. This review aims to systematically explore state-of-the-art RL methods specifically designed for TSC problems, revealing how these tailored approaches better address the complex interactions and challenges inherent in TSC.

4. Advances in reinforcement learning methods for traffic signal control

This section provides a comprehensive review of advanced RL methods applied to TSC in multi-agent environments. First, we explore machine learning-enhanced approaches that tackle key challenges in RL for TSC, such as feature representation limitations, adaptability, generalization, and robustness. Although graph neural networks are a type of machine learning, their suitability for network-level TSC warrants a separate, detailed discussion in Section 4.1. Other machine learning techniques are grouped in Section 4.2. We also cover advanced RL methods like multi-agent coordination mechanisms and multi-objective RL, which are crucial for addressing collaboration and optimization in TSC but are not primarily focused on machine learning. These methods are discussed in dedicated sections due to their distinct role in enhancing RL for TSC. Additionally, emerging AI technologies are introduced separately in Section 4.5. Figure 7 provides an overview of how these advanced RL methods for TSC are categorized.

4.1. Graph neural networks for RL-TSC

4.1.1. Graph message aggregation

In RL-TSC, GNNs enhance the information sharing between signal controllers through message aggregation mechanism, thereby improving the coordination ability between multiple intersections. Typically, intersections or lanes are modeled as nodes to construct graph structures, allowing GNNs to automatically extract features that capture the relationships

between different intersections and dynamically adjust signal control strategies, thereby achieving superior performance in complex traffic environments. Specifically, GNNs aggregate information from neighboring nodes, enabling each traffic signal controller to make decisions based on both its own traffic state and the traffic states of surrounding intersections, thus achieving region-aware collaborative control (Wei, 2019).

Moreover, the integration of GNNs with MARL enhances scalability and learning efficiency in decentralized control architectures. For instance, (Wang 2022a) proposed a region-aware cooperative strategy based on Advantage Actor-Critic (A2C) and GNNs, where local agents cooperate using GNN-enhanced observations from neighboring intersections. This model leverages GNNs to dynamically learn and integrate the spatial influences of surrounding intersections, thus improving the ability of RL agents to reduce vehicle queue lengths and waiting times in multi-intersection traffic grids. Similarly, Zhang (2024c) developed a decentralized multi-agent reinforcement learning model that improves the learning capacity of agents by incorporating GNNs-derived topological information into a MARL framework, which is particularly effective in large-scale road networks. The combination of GNNs and RL not only enhances the model's scalability but also enables more adaptive signal control in dynamic traffic environments.

Although graph message aggregation performs well in improving spatial information sharing efficiency, when combined with reinforcement learning, it can further enhance the system's self-learning ability, automatically adjust signal control strategies, and achieve better signal control effects. However, this approach primarily focuses on spatial information aggregation, neglecting temporal dynamics. As vehicles from neighboring intersections arrive at a later time, this lack of temporal consideration may limit the method's ability to anticipate future traffic flows, potentially constraining the proactive adjustment of signal control strategies.

4.1.2. Spatio-temporal dependency modeling

Compared to graph message aggregation, which primarily captures spatial correlations among intersections, spatio-temporal graph neural networks (ST-GNNs) model both spatial and temporal dimensions of traffic flow, providing a comprehensive representation of spatio-temporal dependencies in TSC. Fundamentally, ST-GNNs integrate temporal information through several common mechanisms. Many architectures combine graph neural networks (for spatial processing) with sequential modeling components like recurrent neural networks (RNNs, e.g. LSTMs or GRUs) or temporal convolutional networks (TCNs). These components explicitly process the sequence of graph states or node features over time, capturing temporal patterns and dependencies. Other approaches utilize attention mechanisms that operate across both spatial and temporal dimensions, allowing the model to dynamically weigh the importance of historical states. When integrated with RL, ST-GNNs leverage these capabilities to not only extract spatial relationships across multiple intersections but also track the evolving dynamics of traffic states over time, facilitating more effective traffic signal scheduling. By incorporating both historical and real-time traffic data processed through these temporal modeling components, ST-GNNs excel at capturing short-term temporal variations, which enhances the adaptability of RL algorithms in complex, dynamic traffic environments. In particular, the ability of ST-GNNs to model spatio-temporal dependencies significantly improves decision-making

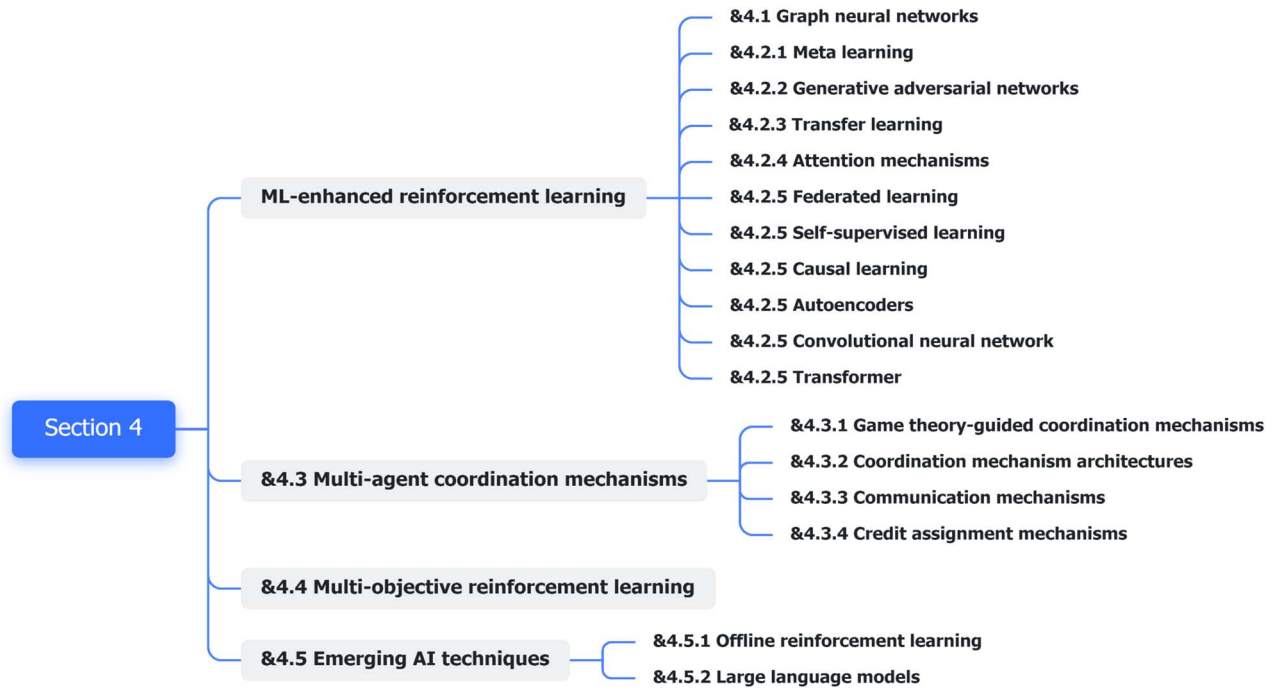


Figure 7. Overview of categorization of advanced RL methods for TSC.

accuracy and the robustness of RL-based strategies to fluctuating traffic conditions, making them a superior choice for real-time TSC.

Several studies have leveraged ST-GNNs to capture spatio-temporal dependencies across intersections. For example, Wang (2022c) exemplifies the integration of dedicated temporal components by combining graph attention networks (GAT) and long short-term memory (LSTM) units within a DQN framework. In their model, GAT is specifically employed to model spatial dependencies through an attention mechanism, while the LSTM component explicitly processes historical traffic data sequences to capture temporal dependencies. This integration enables the distributed optimization of each traffic signal's control, improving coordination across intersections and enhancing overall traffic management efficiency. Similarly, Wang and Liang (2024) employed a MARL framework, incorporating a spatial-temporal attention mechanism that allows agents to dynamically weigh information from different spatial locations and historical time steps, enabling agents to dynamically perceive traffic conditions and coordinate signal control decisions across multiple intersections. The performance in both simulation and real-world traffic scenarios demonstrates the effectiveness of integrating spatio-temporal information to improve traffic flow and reduce congestion.

Other studies have introduced various innovations for modeling spatio-temporal dependencies. For example, Bie et al. (2024) introduced a multi-agent deep reinforcement learning framework based on spatio-temporal graph attention networks (STGATs), optimized for heterogeneous intersections. While details vary, such STGATs typically extend standard graph attention to consider temporal correlations, often by incorporating temporal embeddings or applying attention across time-series data associated with nodes. By incorporating MARL for cooperative signal control and a heterogeneous correlation index to account for intersection geometry and traffic demands, the model enhances signal decisions with fine granularity, reducing delays

and improving vehicle throughput across diverse road networks. Some approaches focus temporal awareness differently: Lin (2024) integrated a graph convolutional network with DQN but addressed temporal dynamics primarily through innovative reward mechanisms: "time-aware pressure of incoming lanes" and "time-aware action consistency," rather than embedding temporal processing directly into the GNN architecture itself. This method, utilizing DQN, still improved decision-making and achieved a 9.5% reduction in average travel time. Meanwhile, Jin (2024) proposed a model combining DQN and GCNs with a multi-head attention mechanism primarily focused on improving signal coordination spatially across intersections. This model effectively captures spatial interactions and managing complex traffic states in a lightweight manner.

ST-GNNs have been employed to address the challenges of large-scale traffic networks and complex traffic flow control. For instance, Kim and Sohn (2022) integrated an asynchronous update mechanism with a deep graph Q-network to model spatio-temporal dependencies across extensive traffic networks, effectively reducing traffic delays.

In summary, the introduction of ST-GNNs, often leveraging components like RNNs, TCNs, or spatio-temporal attention to explicitly model temporal sequences and dependencies, has improved model adaptability to complex traffic environments, enabling TSC systems to better coordinate signals across spatial and temporal dimensions, and enhancing the collaborative efficiency of multi-agent systems. However, incorporating spatio-temporal information significantly increases the model's complexity, leading to higher computational demands due to the additional temporal processing layers or steps, which may limit its applicability in large-scale traffic networks. Additionally, the training process of ST-GNNs can become less stable due to the complexity of modeling these intertwined spatio-temporal dependencies, especially when dealing with noisy or sparse traffic data, potentially slowing convergence and impacting training stability.

4.1.3. Enhancing intersection representation through GNNs

Research on reinforcement learning-based TSC using graph networks has often treated entire intersections or traffic signals as single nodes in the graph. However, some studies have proposed innovative approaches to enhance intersection representations. For instance, Devailly et al. (2022) modeled lanes and vehicles within intersections as nodes. Leveraging graph convolutional networks, this approach adapts to any road network structure, refining the representation of dynamic demands at both the lane and vehicle levels. Such a fine-grained representation improves the model's transferability and scalability. Similarly, different types of traffic entities within intersections (such as roads, vehicles, and traffic signals) have been represented as nodes in the graph (Yang, 2021; Yang and Yang, 2022). These methods utilized heterogeneous graph networks to effectively capture the complex relationships between various node types in the road network, thereby more accurately modeling the interactions among multiple entities in urban traffic systems.

Yoon (2021) modeled lanes within intersections as nodes and used GNNs to represent the traffic state. A distinctive feature of this study is its ability to learn the spatial correlations between lanes, demonstrating strong adaptability and extensibility to previously unseen traffic conditions. Wang (2024b) conceptualized the traffic environment in two distinct layers: the intersection-level and the traffic network-level. At the intersection level, a graph attention network aggregates lane information to elucidate mutual influences among lanes. At the network level, GNNs aggregate information from adjacent intersections, allowing intersection agents to gather comprehensive environmental data.

Overall, unlike earlier approaches that treat graph nodes as a single type, these studies enhance the representation of complex intersection states by modeling multiple traffic entities and control elements as nodes. This approach not only improves the precise modeling of various traffic components within the road network but also enhances the model's generalization capabilities in complex urban traffic environments.

4.1.4. Adaptive network partitioning

Ma and Wu (2024) proposed an adaptive network partitioning method that combined GNNs with Monte Carlo Tree Search (MCTS) to optimize multi-intersection TSC. This method integrated GNNs to dynamically capture spatial dependencies between intersections and utilized MCTS to refine the partitioning process, thereby addressing the limitations of fixed partitioning approaches in adapting to frequently changing traffic patterns. To further enhance this method, a variant of the Qmix algorithm was introduced to combine RL with GNNs, allowing for dynamic region-specific control strategies. This integration allowed the model to learn more effective policies by leveraging both local and global traffic information, resulting in significant performance improvements over state-of-the-art benchmarks, particularly in terms of average travel time and queue length. However, the challenge of training GNNs on large-scale networks persisted, sometimes resulting in suboptimal partitions. While MCTS offered better partitions through explicit search, it increased computational complexity.

4.2. Hybrid machine learning methods

In this section, we provide a comprehensive overview of various machine learning methods integrated with RL, which offer

innovative solutions for TSC, surpassing the capabilities of standalone models. Beyond the extensively discussed GNNs, a range of other machine learning techniques have been employed to enhance RL's performance and adaptability. These approaches have advanced the field of TSC by improving data efficiency, enhancing model generalization, and effectively addressing complex and dynamic traffic environments.

4.2.1. Meta learning

In the context of RL-based TSC, meta-learning enhances algorithmic generalization and training efficiency by quickly adapting to new task scenarios. The key mechanism of meta-learning is to learn shared knowledge across various tasks (e.g., saturated and unsaturated traffic flows) during the meta-training phase, facilitating rapid adaptation to new tasks in the meta-testing phase. For instance, Kim et al. (2023) extracted meta-knowledge from existing tasks, enabling the model to autonomously switch control objectives without relying on external rules or pre-defined models. By utilizing contextual variables to assess current traffic conditions and applying corresponding reward functions, the model swiftly adjusts its strategy to maintain stability in fluctuating traffic environments. These approaches aim to leverage limited sample data for rapid learning, enhancing adaptability to diverse traffic flow settings (Zang, 2020; Zhang, 2020) and significantly reducing training time in new traffic networks (Zhu 2023a). This is particularly crucial for real-world applications, as it overcomes the limitations of traditional deep reinforcement learning, which requires extensive training data and prolonged durations, thereby enabling more effective traffic congestion management during deployment (Lou et al., 2022; Ren, 2022).

4.2.2. Generative adversarial networks

The application of generative adversarial networks (GANs) in reinforcement learning-based TSC focuses on two main areas: addressing data deficiencies and enhancing decision-making quality and stability. GANs enhance reinforcement learning by generating realistic data or features, thereby improving model expressiveness and robustness. For instance, Xu (2024) proposed the use of Wasserstein GANs (WGANs) to estimate missing state spaces in multi-intersection scenarios. Given the potential incompleteness of sensor data from multiple intersections, WGANs were used to construct a comprehensive state space, enabling the reinforcement learning model to function effectively despite data gaps. Similarly, Wang (2022d) developed a decentralized reinforcement learning framework where GANs recovered missing traffic data to overcome limitations caused by restricted communication bandwidth. To improve decision-making quality in TSC, Chen (2023) utilized GANs to infer behavioral trends across multiple agents and, through adversarial training, generated more universally applicable reward functions. This approach enhanced adaptability in diverse traffic scenarios, yet its training process became unstable, especially when balancing complex reward structures across different environments.

4.2.3. Transfer learning

The core concept of transfer learning is to transfer knowledge or strategies learned from a source task to a target task, thereby reducing training time or enhancing performance in the target task. In TSC, transfer learning is primarily used to expedite the model's learning in new environments and reduce its reliance on collecting entirely new data. Norouzi et al. (2021) collected experiences from multiple source agents and classified them before transferring to other agents, enabling the target agent to quickly

adapt to new environments and optimize its decision-making. Furthermore, Da (2023a) narrowed the gap between simulated and real-world environments by employing dynamic action transformation and uncertainty evaluation, enhancing model generalization and real-world performance. These transfer learning techniques allow reinforcement learning models to adapt to complex traffic situations and mitigate performance degradation during real-world deployment. Ge (2022) introduced a multi-view encoder for transfer learning, enhancing agent adaptability in intricate traffic scenarios. This method excels in managing complex multi-intersection scenarios, achieving better traffic flow and reduced delays compared to other models. However, it requires substantial computational resources due to its multi-stage training process, which may pose challenges when applied to larger networks.

Curriculum learning addresses the "curse of dimensionality" in TSC, where the state and action spaces expand significantly with an increasing number of intersections, making it challenging for traditional reinforcement learning methods to efficiently learn optimal policies. In curriculum learning, models first acquire foundational strategies through simpler tasks and then optimize their policies in more complex tasks by overloading policies and reusing experiences. This approach accelerates model convergence and significantly enhances the performance of large-scale TSC (Li et al. 2023b). Liu (2023a) and Liu (2023b) adopted a teacher-student framework, where the teacher guided the student using an importance function to refine its actions. This method simplified model complexity and improved stability, outperforming traditional RL methods in various experiments. Both approaches demonstrate accelerated convergence and improved performance, though their effectiveness may vary with task similarity and the alignment between teacher expertise and student needs. Jiang (2024c) used curriculum learning alongside behavior cloning, guided by industrial traffic control solutions, to ensure practical applicability and adherence to operational constraints, demonstrating a way to integrate expert knowledge directly into the learning process for better transferability to real-world systems.

4.2.4. Attention mechanisms

In reinforcement learning-based TSC, attention mechanisms are widely used to facilitate collaboration within multi-agent systems. Several studies assign attention weights among neighboring agents to enhance their coordination and communication. Local attention mechanisms allow agents to communicate and collaborate effectively in large-scale networks, achieving notable performance in multi-scale road network problems (Gao et al., 2021; Zhu 2023c). Additionally, attention-based interaction mechanisms aggregate critical information at appropriate moments, optimizing the decision-making efficiency of multi-agent systems (Itaya, 2023). The combination of attention mechanisms with multi-agent reinforcement learning in extensive traffic networks effectively reduces computational complexity while enhancing the stability and scalability of control systems (Han 2024b). Yan (2023) introduced a model that combines self-attention mechanisms, thereby augmenting the coordination and information exchange capabilities of traffic signals. The use of self-attention mechanisms enables the model to perceive traffic dynamics across a broader area, resulting in more intelligent signal adjustments.

In addition to coordination, attention mechanisms can also enhance feature representation. For example, Oroojlooy (2020) proposed two attention modules to capture both road-level and

phase-level features. This design enables the model to flexibly adapt to different intersection structures and configurations. Zhao (2024) employed a node-level attention mechanism to dynamically calculate the importance weights between the central node and its neighbors, allowing the model to identify each node's contribution to TSC within the same relationship. A semantic-level attention mechanism further aggregates the impact of multiple relationships on the central node by weighting information across different relationships, thereby enhancing the model's feature extraction and decision-making capabilities.

4.2.5. Others

Federated learning allows agents at each intersection to train locally and share model parameters rather than data, preserving privacy and reducing communication costs. This enables each intersection controller to independently learn TSC strategies while benefiting from collaborative control. Federated learning allows reinforcement learning models to share global knowledge across intersections while adapting to local characteristics and variations. This approach enhances the scalability of TSC, demonstrating higher efficiency and robustness, especially in large-scale and heterogeneous intersection networks (Bao, 2023; Fu and Di, 2023).

Self-supervised learning is used to extract latent features from traffic networks, particularly in scenarios without explicit supervisory signals, by maximizing mutual information to strengthen correlations between different nodes. To ensure training stability, self-supervised learning maximizes mutual information to maintain the stability and consistency of feature embeddings, preventing unstable learning in complex, dynamic environments. As a result, self-supervised models effectively capture intricate structures and dynamic changes within traffic networks, leading to more effective TSC strategies (Yang and Yang, 2022; Liu 2023b).

Causal learning models infer latent causal variables to capture both task-level and time-step-level non-stationary features, which contribute to generating effective TSC strategies. By optimizing the objective function, this causal inference approach enables multi-agent reinforcement learning to better adapt to non-stationary traffic conditions, resulting in improved control outcomes (Yang, 2023). This method shows strengths in achieving stability and adaptability, particularly in transferring knowledge to real-world environments, although its computational complexity could pose challenges in dynamic settings.

Autoencoders compress locally observed traffic state data to dynamically generate concise communication messages, eliminating the need for predefined communication rules and reducing bandwidth usage. By focusing on key dynamic traffic features, they minimize interference from uncertain states and address partial observability issues (Zhu 2023b). The experimental results demonstrated that this approach outperforms traditional communication methods, particularly in reducing communication overhead and adapting to varying traffic patterns. However, while it significantly improves coordination among intersections, the method's performance might decline in scenarios with extremely sparse or dense traffic, where the encoded features might not fully capture sudden changes. Thus, it is most suitable for environments with moderately variable traffic conditions, where the advantages of adaptive message generation can be fully realized.

Cao et al. (2022b) collected vehicle navigation data from traffic networks to compute traffic state matrices, capturing vehicle distributions across road segments and movement trends. Using a convolutional neural network, they extracted features from these spatiotemporal matrices to generate optimal TSC decisions.

Table 1. Summary of hybrid machine learning methods integrated with reinforcement learning. (Adaptability focuses on the capacity to quickly adjust to new scenarios, robustness concerns the resistance to errors or changes, and generalization measures the ability to handle unknown data.)

Machine learning methods	Functionality
GNNs	Coordination (Wei, 2019; Kim and Sohn, 2022; Wang 2022a,c; Bie et al., 2024; Jin, 2024; Lin, 2024; Ma and Wu, 2024; Wang and Liang, 2024; Zhang 2024c); Scalability (Wang 2022a; Zhang 2024c); Enhanced feature representation and adaptability (Yang, 2021; Yoon, 2021; Devailly et al., 2022; Yang and Yang, 2022; Wang 2024b)
Meta learning	Adaptability and generalization (Zang, 2020; Zhang, 2020; Lou et al., 2022; Ren, 2022; Kim et al., 2023; Zhu 2023a)
GANs	Robustness (Chen, 2023; Xu, 2024); Adaptability (Chen, 2023)
Transfer learning	Adaptability (Norouzi et al., 2021; Ge, 2022), Generalization (Da 2023a; Liu 2023a,b; Jiang 2024c)
Attention mechanisms	Coordination (Gao et al., 2021; Zhu 2023c); Scalability and robustness (Han 2024b); Enhanced feature representation (Oroojlooy, 2020; Zhao, 2024)
Federated learning	Scalability and privacy-preserving (Bao, 2023; Fu and Di, 2023)
Self-supervised learning	Enhanced feature representation and generalization (Yang and Yang, 2022; Liu 2023b)
Causal learning	Enhanced feature representation (Yang, 2023)
Autoencoder	Enhanced feature representation and robustness (Zhu 2023b)
Convolutional neural network	Enhanced feature representation (Cao et al. 2022b)
Transformer	Coordination (Zhang et al. 2024b); Generalization (Jiang 2024b)

This method improved traffic flow, particularly in reducing queue lengths and waiting times. However, its reliance on navigation data could limit performance when such data is sparse, though it remains effective with partial data availability.

Zhang et al. (2024b) leveraged the sequence modeling capabilities of Transformer to convert multi-intersection signal control into a sequential decision-making problem. This enabled sequential updates of multi-agent interactions (Grudzien et al., 2022), improving learning efficiency and reducing the complexity of joint policy optimization. Jiang (2024b) introduced a dual-layer Transformer structure where the lower layer facilitates coordination among intersections, and the upper layer enhances cross-city generalization. This design demonstrated strong transfer performance, consistently outperforming existing methods in various challenging scenarios. While it effectively improves adaptability and coordination, its reliance on extensive training data across diverse scenarios may pose challenges in data-limited contexts.

We have summarized the role of machine learning methods in RL-TSC, as shown in Table 1.

4.3. Multi-agent coordination mechanisms

Building upon the various machine learning methods integrated with RL discussed previously, this section delves specifically into the mechanisms and frameworks explicitly designed to manage coordination among multiple agents (intersections) in TSC. While techniques like attention mechanisms (Section 4.2.4) or specialized communication encodings (Section 4.2.5) can serve as components enabling coordination, the focus here is on the overarching strategies and dedicated approaches developed within the MARL field for TSC. As urban traffic networks become increasingly complex, collaboration among multiple agents at various intersections is crucial. Achieving effective coordination typically involves addressing several interrelated challenges: establishing a structural framework for interaction (coordination architectures), enabling agents to share relevant information through communication mechanisms, and developing methods for agents to learn their roles and contributions toward the collective goal via credit assignment. Game theory (discussed first in Section 4.3.1) provides a powerful theoretical foundation for analyzing and designing these coordination strategies. We explore these key components. Together, these elements aim to optimize signal

strategies and enhance overall traffic flow. We summarize the related works on cooperation mechanisms in RL-TSC in Table 2.

4.3.1. Game theory-guided coordination mechanisms

The integration of game theory into MARL for TSC provides a robust framework for optimizing cooperation among agents. Game-theoretic methods, such as Nash equilibrium and the mean field method, offer structured approaches to address the challenges of multi-agent coordination by balancing the strategies of agents across multiple intersections. For example, Nash equilibrium defines a stable state where each agent adopts a strategy that optimizes its local performance objective (its 'individual' goal within the control setup) as the best response to the strategies of others, such that no agent benefits from unilaterally deviating. Establishing this equilibrium among the control agents is a primary goal of game-theoretic MARL, as the resulting stable coordination is expected to enhance overall network efficiency compared to uncoordinated control. Shen (2023) proposed a hierarchical game model based on Nash equilibrium, specifically utilizing models like the Nash-Stackelberg model. The model partitions the traffic network into a primary layer (key intersections) and a subordinate layer (secondary intersections). The primary intersections coordinate signal control strategies using Nash equilibrium, while the subordinate intersections respond optimally to the strategies of the primary layer. This hierarchical structure effectively reduces system complexity, enhancing the efficiency of TSC. In terms of algorithm design, MARL algorithms that integrate Nash equilibrium, such as Nash Q-learning (Wu, 2022) and Off-policy Nash DQN (Zhang 2023a), take into account the states and strategies of neighboring agents. These algorithms employ quasi-game techniques at each timestep to find Nash equilibrium points. They then update each agent's Q-values to identify the optimal strategy combination for the multi-agent system. In this manner, agents share behavioral information during interactions, achieving Nash equilibrium and ensuring the convergence and stability of the overall strategy. Yan and Wang (2024) utilized mixed-strategy Nash equilibrium to optimize decision-making. The use of multi-layer gated recurrent unit networks facilitates the sharing of action information among neighboring agents, mitigating instability in multi-agent environments and accelerating convergence in large-scale traffic networks.

Table 2. Summary of cooperation mechanisms in RL-TSC.

Coordination mechanism type	Key methods	Main characteristics	Related Works
Game theory-guided coordination mechanisms	Nash equilibrium	- Optimizes individual and overall performance - Improves stability and convergence	Wu (2022); Zhang (2023a); Yan and Wang (2024) Wang (2021); Hu (2023a); Zhang (2023b) Shen (2023)
	Mean field method	- Reduces dimensional complexity - Simplifies decision-making in large-scale networks	
	Hierarchical game model	- Layered network structure - Reduces system complexity	
Coordination mechanism architectures	Network partitioning methods	- Decomposes networks into manageable regions - Enhances control efficiency in large-scale networks	Jiang (2022a); Zhu (2024); Gu (2024a) Xin (2023); Ren (2024a)
	Hierarchical coordination	- Uses skill discovery and tiered structures - Balances local cooperation and global performance	
Communication mechanisms	Strategic communication	- Predicts agent behaviors and adapts to needs - Communicates selectively to reduce overhead	Bokade et al. (2023); Zhou (2024) Zhang (2022a); Guo (2023b)
	Context-aware communication	- Emphasizes critical traffic data - Incorporates supply-demand modeling	
Credit assignment mechanisms	Counterfactual evaluation	- Compares actual actions with hypothetical alternatives - Provides accurate measurement of contribution	Goel (2023); Song et al. (2024) Li (2024b); Zhang and Zhou (2025) Wang (2022b)
	Value decomposition methods	- Decomposes global value function to guide local decisions - Dynamically adjusts weights based on local rewards	
	Differential reward-based approaches	- Isolates influence of neighboring agents - Measures independent contributions	

The mean field method simplifies interactions in large-scale networks by approximating agent behavior as interactions with an 'average' agent. These techniques allow MARL-TSC systems to achieve stable, convergent strategies, mitigating the complexity inherent in dynamic, large-scale traffic networks (Liu, 2021). Additionally, hierarchical game models and other game-theoretic approaches reduce system complexity and improve scalability, making game theory a key tool for optimizing multi-agent TSC.

In reinforcement learning, the mean field method plays a pivotal role in MARL for RL-TSC, primarily by reducing dimensionality and simplifying decision-making. In large-scale traffic networks, multiple agents, such as traffic lights at different intersections, must collaborate to manage control. However, as the number of agents increases, the joint action space grows exponentially, leading to a dimensionality explosion and convergence difficulties. The mean field method addresses this issue by transforming each agent's joint action into interactions between the agent and the average action of its neighboring agents. This effectively converts a multi-agent reinforcement learning problem into one of interaction between a single agent and an 'average agent,' significantly reducing the dimensionality and complexity of the action space (Wang, 2021; Hu 2023a). Experimental results from Zhang (2023b) demonstrated that the mean field method outperforms traditional deep reinforcement learning algorithms in regional TSC, achieving higher average vehicle speeds, shorter waiting times, and more stable reward convergence. By decomposing complex multi-agent interactions, the mean field method markedly enhances the efficiency and effectiveness of regional TSC compared to fixed-timing strategies and other deep reinforcement learning algorithms, showcasing its potential advantages in large-scale multi-agent TSC scenarios.

4.3.2. Coordination mechanism architectures

Effective multi-agent coordination first requires defining the structural framework within which agents interact. In the field

of RL-TSC, coordination mechanism architectures determine how agents are organized or how the network control problem is decomposed to facilitate collaboration, optimize traffic flow, and reduce congestion. Graph partitioning-based approaches are particularly effective, as they decompose large-scale traffic networks into sub-regions for individual or joint optimization, enhancing the efficiency of control over the entire network.

Jiang (2022a) proposed a network decomposition strategy using the level of connectivity, defined as the average residual capacity between connected intersections, to partition the network into closely connected subgraphs. Gu (2024a) introduced a constrained network partitioning method that divides the network into star-topology regions, each consisting of a central node and multiple leaf nodes. This design reduces signal conflicts among intersections within each region, improving regional control. Xin (2023) proposed a hierarchical structure for intersection coordination using skill discovery, with a cooperation policy learning probabilities for collaboration and an execution policy implementing signal control based on these probabilities. This mechanism enhances efficient coordination under varying traffic conditions. Zhu (2024) used coordination graphs to model paired relationships between intersections, which are dynamically updated based on traffic demand. Independent intersections apply RL strategies individually, while paired ones are jointly controlled using the max-plus algorithm to balance traffic flow and respond quickly to demand changes. Ren (2024a) developed a two-tier coordination mechanism: local cooperation uses a local cooperation factor to balance rewards between intersections and their neighbors, and global coordination optimizes the network-wide total reward to update the local cooperation factor. This design balances local cooperation and global performance, improving multi-intersection TSC. While traditional methods often assumed static structures with fixed neighbors, recent work like CoSLight (Ruan, 2024) introduced dynamic collaborator selection. CoSLight co-optimized signal control with a policy that adapted in real-time, enabling flexible coordination that responded to changing traffic patterns.

This marked a significant advancement in context-aware multi-agent coordination for TSC.

4.3.3. Communication mechanisms

Building upon the coordination architectures, communication mechanisms are critical in multi-agent reinforcement learning for TSC, facilitating the necessary information sharing among agents within the established structure. With communication mechanisms, agents can access traffic state information from neighboring nodes or other agents within the network, enabling more informed decision-making, congestion alleviation, and improved traffic flow efficiency. Furthermore, well-designed communication mechanisms help reduce information noise, minimize communication overhead, and enhance the system's stability and robustness (Li, 2021).

Zhou (2024) proposed an incentive-based communication mechanism consisting of a teammate module and a communication module. The teammate module predicts other agents' behaviors by generating customized messages from time-series data. Meanwhile, the communication module uses an attention mechanism to generate tailored communication information and calculate communication weights, creating personalized messages for different agents. Bokade et al. (2023) developed a selective communication framework that allows agents to determine 'when' and 'with whom' to communicate. By learning communication strategies, agents communicate only when necessary and selectively transmit relevant information to neighboring agents, thereby reducing channel usage and improving overall control efficiency. To highlight crucial information in multi-intersection control, Zhang (2022a) introduced a neighborhood relevance-based communication strategy. By calculating correlations between neighboring agents' traffic states, agents can weight communication information to emphasize critical traffic data while suppressing less relevant information, making it particularly effective in scenarios where intersections differ in importance and traffic volume. Addressing the impact of signal synchronization on traffic control, Guo (2023b) proposed a method that reduces unnecessary communication overhead through active communication and action correction. This method also incorporates supply-demand modeling to balance traffic demand with road capacity, ensuring timely communication and enhancing the regulation of traffic flow.

4.3.4. Credit assignment mechanisms

Once agents are organized within an architecture (Section 4.3.2) and potentially sharing information through communication (Section 4.3.3), credit assignment plays a crucial role in enabling them to learn effective cooperation and coordination strategies to achieve global objectives. Given that each agent's actions within a complex traffic network impact overall traffic flow, accurately assessing and allocating each agent's contribution is essential for learning. Credit assignment enables agents to understand their influence and learn to coordinate and collaborate more effectively, optimizing the overall network performance rather than focusing solely on individual rewards. Moreover, it improves learning efficiency, ensures model convergence, and strengthens agent collaboration in large-scale, complex TSC tasks.

Song et al. (2024) proposed a counterfactual multi-agent deep actor-critic framework, featuring a centralized critic network for evaluating joint strategies. Counterfactual estimation allocates credit by comparing an agent's actual actions with hypothetical alternatives, providing a more accurate measure of its contribution to the overall objective. Wang (2022b) introduced a

Table 3. Summary of multi-objective in Section 4.4.

Reference	Multi-objective
Li (2022)	Fuel economy, safety, and travel time
Wu (2023b)	Fuel economy and speed
Jin and Ma (2019)	Delay, environmental sustainability and fuel economy
Jamil and Nower (2022)	Travel time, waiting time, and halting number
Saiki and Arai (2023)	Travel time, driving safety, and throughput
Zhang (2024a)	Travel time, driving safety, and environmental sustainability
Fang (2023b)	Queue lengths, driving safety, and network coordination

differential reward-based approach for credit assignment in large-scale TSC. By isolating the influence of neighboring agents, differential rewards measure independent contributions, accelerating policy convergence, and improving overall control performance. Li (2024b) utilized value decomposition to derive globally optimal strategies. A high-level agent guides decision-making updates of lower-level agents, ensuring effective coordination across intersections of varying types. Zhang and Zhou (2025) proposed an adaptive value decomposition-based multi-agent actor-critic approach for TSC, where credit assignment is achieved through dynamically adjusting weight coefficients based on local rewards, enabling agents to adaptively coordinate and optimize the overall traffic network flow. Goel (2023) achieved network-level coordination by estimating each agent's marginal contribution to its local neighborhood through a distributed approach. A locally centralized critic network employed counterfactual reasoning to compute these contributions, addressing multi-agent cooperation challenges. Zhang et al. (2024b) modeled multi-intersection TSC as a sequential decision-making problem (Grudzien et al., 2022), addressing credit assignment through imitation learning and leveraging Transformers for sequential updates. This approach improves learning efficiency and reduces the complexity of optimizing joint policy.

4.4. Multi-objective reinforcement learning

In urban traffic management, signal control necessitates balancing multiple objectives, often encompassing traffic efficiency (reducing waiting times and queue lengths), traffic safety (minimizing traffic conflicts), and environmental benefits (lowering carbon emissions and fuel consumption). Unlike traditional single-objective optimization problems, multi-objective signal control focuses on balancing these sometimes conflicting or independent goals to achieve a more balanced traffic performance. However, trade-offs may arise between different objectives, where optimizing one goal could negatively impact others. Multi-objective reinforcement learning seeks to optimize these diverse goals either by assigning weights to each objective at each decision-making state or by directly designing multi-objective reward functions. Nonetheless, the priorities of different objectives may vary with time, location, and traffic flow, which significantly increases the complexity of the approach. As a result, dynamic strategy adjustments are often needed to adapt to real-time traffic conditions. The multi-objective summaries of the studies discussed in this section are presented in Table 3.

One approach to addressing multi-objective problems in TSC is to design a multi-objective reward function, achieving comprehensive optimization through dynamic weight adjustment to balance various objectives (Fang 2023b; Wu 2023b).

For example, [Jamil and Nower \(2022\)](#) proposed dynamically adjusting the weights of different objectives based on real-time traffic indicators, such as vehicle waiting times and travel speeds, to optimize signal control strategies. Similarly, [Li \(2022\)](#) integrated traffic signals, fuel efficiency, and safety into a unified reward function, using a potential shaping function to guide the agent toward optimal strategies across diverse scenarios, with weights adjusted according to real-time traffic conditions for comprehensive optimization. [Zhang \(2024a\)](#) proposed a dynamic weight adjustment method for multi-objective reward functions, where initial weights are based on expert opinions, public preferences, or policy guidelines and then adjusted according to real-time traffic conditions. This approach allows for prioritizing safety when traffic conflicts increase, while efficiency is emphasized during prolonged waiting times. The method effectively balances safety and efficiency, achieving a significant reduction in traffic conflicts. However, the reliance on accurate real-time data and the complexity of adjusting weights for different scenarios may limit its applicability in environments with limited data availability.

A common approach to simplify multi-objective problems is to transform them into single-objective ones. [Jamil et al. \(2020\)](#) introduced a Dynamic Weights Multi-Objective Reward Architecture (DWMORA), which adjusts reward weights based on real-time traffic conditions. This approach outperforms fixed-weight methods by dynamically adapting to changes in traffic flow, leading to reduced waiting times and improved travel times. However, its performance can be sensitive to the accuracy of weight adjustment, making it less effective when abrupt changes occur in the traffic environment. [Jin and Ma \(2019\)](#) proposed a multi-objective reinforcement learning model based on a constrained Markov decision process, where the highest-priority objective is set as the primary optimization target, and other objectives are treated as constraints. This method effectively balances objectives across varying traffic conditions using a threshold lexicographic ordering method, resulting in improved adaptability. Nevertheless, the complexity of tuning constraints can limit its application in highly dynamic traffic scenarios, requiring careful parameter adjustments.

[Saiki and Arai \(2023\)](#) extended the MDP framework into a multi-objective MDP, deriving Pareto optimal strategies that adapt to different traffic flow ratios. Their method allows for dynamic switching between strategies, ensuring flexibility in response to changing traffic patterns. While it achieves the shortest travel times compared to rule-based and single-objective RL methods, its reliance on flow-specific policies may reduce generalizability when applied to significantly different traffic conditions.

4.5. Emerging AI techniques

4.5.1. Offline reinforcement learning

Offline reinforcement learning (Offline RL) is a method designed for training reinforcement learning agents using pre-collected datasets. In contrast to online reinforcement learning, which requires continuous interaction with the environment, Offline RL operates on a static dataset. This method can be broadly categorized into two main approaches: Bellman equation-based reinforcement learning and sequence modeling-based fully supervised learning. The former approach constrains the policy's deviation from the offline data distribution through approximate dynamic programming techniques (e.g. conservative Q-learning and policy regularization) to ensure both stability and generalizability ([Fujimoto et al., 2019](#); [Kumar, 2020](#)). The latter approach, exemplified by the Decision Transformer, reformulates offline

reinforcement learning as a sequence modeling task, directly predicting optimal actions via supervised learning ([Chen, 2021](#)).

In reinforcement learning-based TSC, Offline RL can be applied to derive policies from historical traffic data. Bellman-based approaches ensure model stability and enhance the capability of Offline RL to handle distributional shifts and uncertainties. For example, [Kunjir \(2023\)](#) introduced a 'pessimistic' reward mechanism within a model-based Offline RL framework, effectively mitigating overestimation issues caused by discrepancies between offline data and real-world conditions. Conversely, sequence modeling approaches provide adaptability to complex traffic environments by utilizing sequence information to optimize signal control policies. Specifically, [Wu \(2023a\)](#) employed gated Transformer networks to input sequences of states, actions, and rewards, enabling the model to directly predict optimal signal control actions without relying on dynamic programming. This supervised sequence modeling demonstrates superior stability and training efficiency, thereby reducing the learning costs for Offline RL models. Offline RL effectively addresses the high exploration costs and safety concerns associated with online interaction processes. Consequently, some studies employ an 'offline-to-online' strategy, leveraging Offline RL for model pretraining, followed by online fine-tuning to enhance adaptability. Both [Ma and Wu \(2023\)](#) and [Sun \(2024\)](#) proposed similar offline-to-online frameworks: pretraining in the offline phase to encompass diverse traffic scenarios, and subsequently adapting and optimizing in the online phase. This approach reduces exploration costs and mitigates risks in real-world environments, while significantly decreasing the time required for online learning.

4.5.2. Large language models

Both traditional methods and RL approaches often suffer from limited generalization capabilities. To address this limitation, LLMs, known for their strong generalization ability and human-like reasoning across diverse tasks, present novel solutions. By integrating LLMs into TSC, they can serve as decision-making agents that emulate human intuition to manage traffic signals more effectively, thereby enhancing control efficiency.

[Lai \(2023\)](#) proposed a methodology for enhancing TSC by exploiting the decision-making capacities of LLMs. This framework actively monitors real-time traffic data, constructs sophisticated prompts, and utilizes 'chain-of-thought' reasoning to simulate human-like decision processes within LLMs. The approach involves training through imitation learning and policy optimization, aiming to improve the efficacy and scalability of signal control strategies in a cost-effective manner. The experimental results indicate that this method provides strong generalization capabilities and interpretability, particularly in diverse traffic scenarios. However, its reliance on prompt design can be a limitation when faced with highly dynamic conditions, requiring further refinements for broader adaptability. [Da \(2024b\)](#) introduced an innovative solution for transitioning RL from simulated environments to real-world applications. By harnessing the knowledge-based reasoning of LLMs, this framework integrates a dynamic modeling module that utilizes prompts to adapt to changes in real-world systems, effectively bridging the performance gap between simulations and reality. The approach demonstrates notable improvements in adapting RL policies under varying environmental factors, such as weather and road conditions. While the method enhances the robustness of RL strategies in practical applications, its success is highly dependent on the quality of the prompts, which may require extensive

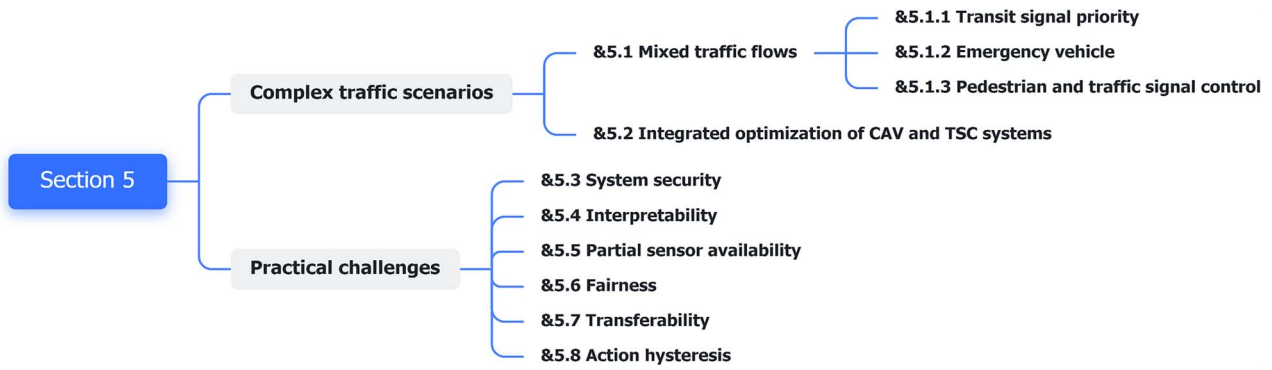


Figure 8. Overview of complex traffic scenarios and practical challenges in Section 5.

domain-specific knowledge for optimal performance. (Yuan et al., 2025) introduced a cooperative LLM agent framework for network-wide TSC. By using spatiotemporal graph representations and complexity-aware reasoning, it enabled coordination among agents at different intersections. With a specialized fine-tuning strategy, it outperformed traditional methods and previous LLM/RL approaches, showcasing the potential of cooperative LLM agents in complex control tasks.

5. Extended research in complex traffic scenarios and practical challenges

This section explores recent research on RL-TSC, addressing both complex traffic scenarios and the practical challenges of model application. Complex traffic scenarios refer to conditions that deviate from idealized settings, such as those assuming uniform vehicle types. Instead, real-world conditions include mixed traffic flows (buses, emergency vehicles, and pedestrians) interacting with traffic signals, as well as the integrated optimization of connected autonomous vehicles and TSC systems. Practical challenges focus on real-world deployment and maintenance, including system security, interpretability, partial sensor availability, promoting fairness in decision-making, transferring models across different traffic networks, and mitigating action hysteresis, where abrupt signal changes risk causing accidents and require the introduction of delays. Figure 8 illustrates these issues in both complex traffic scenarios and practical challenges.

5.1. Mixed traffic flows

RL-TSC in multimodal traffic environments aims to optimize traffic flow across various participants, such as cars, buses, pedestrians, and non-motorized vehicles. Traditional TSC primarily focuses on vehicles, often overlooking the priorities and needs of other transport modes. However, as urban traffic becomes increasingly complex, the coordinated control of diverse traffic types has become crucial. In mixed traffic scenarios with both connected vehicles (CVs) and human vehicles (HVs), CVs provide real-time status information, while HV information must be inferred or is challenging to obtain. Yang and Fan (2024) proposed a state space that combines detailed CVs data (e.g. position, speed) with aggregated sensor data (e.g. lane queue lengths), enabling the controller to leverage CVs data while compensating for HVs information gaps. Wang (2023a) applied deep reinforcement learning to coordinate multiple traffic modes, aiming to enhance social equity by minimizing the average waiting time across all participants. Vlachogiannis (2024) explored multimodal traffic involving single-occupancy vehicles (SOVs),

high-occupancy vehicles (HOVs), microtransit, and public transit. They introduced a human-centric pressure in the reward function, prioritizing signal phases serving more passengers to enhance traffic efficiency per capita by extending green phases for vehicles with higher passenger volumes. The summary of the state and reward settings for various studies under different scenarios in this section is presented in Table 4.

5.1.1. Transit signal priority

Transit signal priority (TSP) is a complex challenge that aims to balance the priority given to transit buses with the flow of other traffic, enhancing the punctuality and speed of public transport. In the context of TSP, RL-TSC must take into account bus arrival times, travel durations, and trade-offs with other vehicles. RL-TSC distinguishes itself from conventional signal optimization by utilizing online, real-time learning of dynamic traffic conditions, thereby optimizing delays and on-time performance for both buses and general traffic—outcomes often unattainable with traditional preset or passive priority controls (Long, 2022). The RL-TSC model for TSP also aims to incorporate specific objectives of bus operations, such as maintaining consistent headways and schedules, thereby improving the efficiency and reliability of transit services.

Yu (2023) employed a MARL framework to integrate TSC with transit priority and waiting time regulation, modeling each intersection as an independent agent making decisions based on real-time traffic and bus information. The state space included variables such as vehicle queue lengths, bus locations, and speeds, all of which were crucial for effective decision-making. The reward function was designed to simultaneously minimize total vehicle waiting time and balance bus headways. Experimental results showed that the approach by Yu (2023) effectively reduced average vehicle waiting times and maintained bus headway consistency. However, this method required higher communication bandwidth and computational resources for managing high-frequency bus routes. The dual-objective TSP algorithm developed by Hu (2023b) excelled in optimizing bus punctuality and travel speed, especially showing high flexibility in handling delayed buses. Nonetheless, it struggled with delayed strategy adjustments in response to frequent traffic fluctuations. Kampitakis et al. (2023) proposed a reinforcement learning algorithm that dynamically adjusted traffic signals and bus lane allocations based on real-time traffic and bus conditions, such as passenger numbers and bus arrival times, to minimize average passenger waiting times and bus delays. This approach demonstrated greater adaptability in integrating dynamic bus lane allocation and signal adjustments, making it suitable for complex traffic scenarios that require balancing

Table 4. Summary of scenarios, states, and rewards in Section 5.1.

Reference	Scenario	State	Reward
Yang and Fan (2024)	Mixed traffic with CVs and non-CVs	Position and speed matrix, feature-based vector (queue length, average speed, cumulative waiting time)	Reduction in average delay
Wang (2023a)	Cars, buses and pedestrians	Position, queue length, waiting time, vehicle speeds	Minimize waiting time, enhance social equity
Vlachogiannis (2024)	Ridesharing and HOV prioritization	Vehicle position, active vehicle range, vehicle occupancy	Minimize person-based pressure, prioritize HOV travel time savings
Yu (2023)	Transit signal priority	Vehicle counts, bus headways, occupancy	Minimize total stop time, balance bus headways
Hu (2023b)	Transit signal priority	Headway deviation, travel time, green phase duration	Reduce headway deviation, minimize travel time
Kampitakis et al. (2023)	Transit signal priority	Passenger counts, bus delay, vehicle queue length, bus lane status	Minimize passenger waiting time, reduce bus delays
Cao et al. (2022a)	Emergency vehicle	EMV location, queue length, waiting time, phase status	Minimize EMV wait time, balance non-EMV delays
Su (2023)	Emergency vehicle	Traffic density, vehicle positions, EMV locations	Reduce EMV travel time, minimize overall traffic delay
Wang (2023b)	Emergency vehicle	Traffic density, EMV status, queue lengths	Maximize throughput, reduce delay for EMV and non-EMV
Zhu (2022)	Mixed pedestrian-vehicle traffic	Vehicle and pedestrian positions, waiting time, traffic context	Minimize pedestrian waiting time, reduce vehicle congestion
Liu et al. (2024)	Mixed pedestrian-vehicle traffic	Vehicle positions, pedestrian locations, traffic density	Minimize vehicle and pedestrian delay, ensure emergency response
Xu (2023b)	Mixed pedestrian-vehicle traffic	Vehicle density, pedestrian density, current signal phase	Minimize travel time for vehicles and pedestrians, improve safety
Yazdani (2023)	Mixed pedestrian-vehicle traffic	Vehicle positions, pedestrian counts, signal status	Minimize total user delays, including vehicle and pedestrian delays

different transportation modes. However, its performance under low communication conditions required further validation.

5.1.2. Emergency vehicle

Emergency vehicle (EMV) signal priority control involves balancing the demands of emergency vehicles and regular traffic in a multimodal environment. Similar to transit signal priority (TSP), the primary objective of EMV signal priority is to minimize the travel time of emergency vehicles to enable rapid response (Shamsi et al., 2022). In contrast to TSP, EMV priority requires consideration of more dynamic, real-time factors, such as coordinating multiple emergency vehicles from various directions, maintaining efficient flow for regular traffic, and the interaction between emergency vehicles and traffic signals in real-time. Additionally, the urgency and unpredictability of EMVs make signal control strategies more complex, necessitating rapid EMV passage with minimal disruption to other vehicles.

Cao et al. (2022a) designed a state space comprising the lane density of the intersection, signal phases, and information on the position and travel direction of EMVs, aiming to enhance decision-making. The reward function seeks to balance EMV priority by minimizing their travel time while mitigating delays for regular traffic, achieving an equilibrium between traffic flow and emergency response. Similarly, Wang (2023b) used a hybrid model that assigns significant weight to EMV priority in the reward function to ensure their expedited passage. Su (2023) applied MARL combined with dynamic path planning to couple EMV route optimization and signal priority in real-time. They developed a pressure-based reward function to minimize travel time for both emergency and regular vehicles. These studies have improved the traffic efficiency of EMVs to varying degrees. However, they also face some common challenges. Firstly, these methods rely

heavily on the precise design of state space and reward functions during practical deployment, which can make it difficult to maintain adaptability in environments with highly uncertain traffic conditions. Secondly, the coordination requirements of multi-agent methods may reduce their effectiveness in scenarios with unreliable communication or in large-scale networks.

5.1.3. Pedestrian and traffic signal control

Recent studies have explored pedestrian-involved TSC using RL to alleviate congestion, enhance traffic flow efficiency, and ensure the safety of both pedestrians and vehicles. Traditional RL-TSC research often focuses on vehicles, overlooking pedestrian dynamics. However, pedestrian flow significantly impacts signal control strategies due to differences in behavior, speed, and interactions with vehicles compared to motorized traffic. Incorporating pedestrians into RL-TSC introduces greater diversity and complexity, such as conflicts between pedestrians and vehicles, pedestrian influence on signal timing, and the need for coordinated control across various traffic streams. Consequently, RL-TSC accounting for pedestrians presents greater challenges in state space design, reward formulation, and control strategy development compared to vehicle-focused TSC.

Zhu (2022) incorporated pedestrian counts and waiting times into the state space and combined both pedestrian and vehicular waiting times in the reward function. They used long short-term memory (LSTM) networks to capture hidden traffic patterns and a context-aware mechanism to integrate neighboring agent states, effectively managing mixed pedestrian and vehicle traffic. The experimental results demonstrated a significant improvement in traffic flow control in multi-intersection scenarios, although the training process was computationally intensive and complex. Yazdani (2023) developed an extended reward function that accounts

for delays caused by interactions between vehicles and pedestrians, including jaywalking incidents. Their model optimizes green time allocation and significantly reduces total user delays. However, the model's sensitivity to parameter settings presents a potential limitation. Xu (2023b) leveraged 6G positioning technology to analyze diverse pedestrian queuing behaviors, integrating pedestrian flow into vehicle signal control schemes and developing an enhanced 'pressure' reward function. Their method outperformed traditional approaches in distributed control across multi-intersection scenarios, but its reliance on 6G technology may limit practical deployment. Liu et al. (2024) integrated real-time pedestrian states, such as location and waiting time, with traffic flow and road environment data (e.g. vehicle density and road topology) to construct a comprehensive state space. The reward function optimizes signal timing by minimizing vehicle and pedestrian delays, and penalizes excessive pedestrian waiting times. This approach effectively enhances pedestrian satisfaction and reduces the risks associated with prolonged waits, though the model's complexity may necessitate advanced computational resources for implementation in intelligent transportation systems.

5.2. Integrated optimization of CAV and TSC systems

In the context of connected and autonomous vehicles (CAV) scenarios, TSC through reinforcement learning encounters novel challenges, mainly centered on the joint optimization of traffic light signals and vehicle behaviors. Compared to traditional TSC, CAV settings demand the integrated optimization of two systems: the real-time adjustment of traffic lights and the speed and trajectory control of CAV queues (Guo and Ma, 2021). This expands the problem beyond merely determining the optimal phase sequence and duration of signals to enhance traffic efficiency. It also necessitates accounting for the dynamic behavior of CAV queues to ensure smoother traffic flow. In this complex multi-agent system, traffic lights determine the right-of-way, while vehicles select optimal speeds and paths based on the state of both signals and other vehicles. First, given that CAVs possess communication and autonomous control capabilities, traffic light control strategies must be efficiently coordinated with vehicle behavior to enable smooth passage through intersections, minimizing wait times and energy consumption (Zhu, 2021; Lombard, 2023). Second, the bidirectional exchange of information between CAVs and traffic lights means that signal control must not only account for current traffic flow but also predict future traffic states and CAV behavior to achieve globally optimal regulation. Furthermore, in mixed traffic environments, the differing behavior patterns of CAVs and human-driven vehicles must be concurrently managed, complicating signal and platoon control. Novel approaches using collaborative LLMs, like CoMAL (Yao, 2024), are emerging to manage interactions specifically in these mixed-autonomy settings, focusing on vehicle-level coordination. In summary, TSC within CAV contexts requires integrated optimization across multiple dimensions—multi-agent coordination, real-time decision-making, and traffic flow prediction—to achieve efficient, safe, and environmentally friendly transportation outcomes (Guo et al. 2023a).

In the integrated optimization of CAV-TSC, managing vehicle queue behavior is a core challenge. CAVs can coordinate with traffic signals through speed guidance—adjusting speed and queueing behaviors—to enable efficient platooning through intersections (Maadi, 2022). Berbar (2022) introduced a 'dual-agent' system, where one agent controls signal phases and the other

governs vehicle speed, ensuring smooth CAV passage through traffic lights while optimizing fuel consumption. Nonetheless, the complexity of coordinating two agents increases the computational demand, which may pose challenges in large-scale networks. Sun (2023) introduced a hierarchical reinforcement learning framework that achieves effective coordination between traffic signals and autonomous vehicles. The multi-agent structure enables detailed control at both macro (signal control) and micro (vehicle speed adjustment) levels, enhancing overall traffic efficiency. However, the hierarchical setup may require extensive training time, especially in highly dynamic traffic environments. Similarly, Li (2024a) introduced a cooperative multi-agent DRL framework that integrates CAV platooning with traffic signal synchronization. It shows a marked improvement in travel times and fuel consumption through dynamic synchronization. Despite its effectiveness in complex urban environments, the need for extensive communication infrastructure may pose a challenge for real-world implementation. Leveraging vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication, Dai (2024b) enabled real-time data exchange between CAVs and traffic signals, using a distributed multi-agent DRL structure where both signal and vehicle platoon agents make local decisions while sharing a global objective, achieving optimal traffic flow across the network. Gao et al. (2024) leveraged arrival time vectors (ATVs) to synchronize traffic signals with CAV platoons, achieving notable reductions in travel delays and fuel consumption. The real-time synchronization is particularly effective for reducing stop times at intersections. However, the approach may struggle with unpredictable human-driven vehicle behaviors, limiting its applicability in mixed traffic conditions. Zhang (2025) proposed a DRL-based cooperative control framework that integrates TSC and CAVs to enhance intersection safety and efficiency. The framework employs a multi-objective reward function and a speed planning model to optimize signal timing and CAV velocities based on traffic flow analysis.

5.3. System security

The issue of system security primarily concerns the following aspects: (1) Privacy protection: The need for real-time collection and processing of vehicular data raises concerns about potential breaches of user privacy; (2) Adversarial attacks: RL models are vulnerable to potential black-box or white-box attacks, which can compromise their performance, result in erroneous decisions, or even cause system paralysis, especially in complex traffic conditions; (3) Robustness: The model may be exposed to anomalies or malicious inputs (such as noise or tampered data) in real-world scenarios, impairing its ability to accurately perceive and respond to traffic states. Addressing these security challenges during system deployment is essential for ensuring the safe, stable, and efficient operation of TSC.

Several studies have proposed various approaches to enhance the security of TSC systems. Muller and Sabatelli (2022) proposed an action-shielding method to ensure that traffic signals manage vehicle flow under conditions of physical safety and psychological comfort, thereby preventing RL policies from making hazardous decisions. Ying (2022) employed multiparty computation and secret sharing techniques to effectively protect vehicle information privacy in deep reinforcement learning, ensuring that no sensitive data is exposed during the signal control process. Ren (2024b,c) explored the robustness of MARL systems against adversarial attacks, designing a black-box multi-objective attack strategy to disrupt signal control policies by manipulating rewards in key states. Xu (2023a) simulated special traffic state anomaly data

(TSAD) and analyzed the vulnerability of DRL models, and proposed a model based on reward detection and neuron value masking to enhance system robustness. Haydari et al. (2021) examined both white-box and black-box adversarial attacks in DRL-based TSC, developing a low-cost integrated anomaly detection model that combines multiple detection algorithms to achieve effective results across various attack scenarios. Xu (2025) proposed an Integrated Structure and Robustness Enhancement framework that combines robust distillation for wrong knowledge suppression at the macro level and structure compression for robustness enhancement at the micro level, achieving both model compression and robustness improvement against abnormal traffic data.

5.4. Interpretability

In the practical deployment of RL for TSC, interpretability becomes a crucial concern. RL models are often built using deep neural networks, leading to a 'black-box' phenomenon that obscures their decision-making process. In the context of TSC, understanding the decision logic of these models is essential for ensuring reliability and transparency, especially in high-risk scenarios where safety and stability are critical. Interpretability not only fosters trust in the model but also enables experts to assess and validate its behavior, thereby enhancing transparency in real-world applications. Therefore, exploring methods to improve the interpretability of RL models is crucial for their practical deployment in TSC systems.

Several studies have proposed interpretable reinforcement learning models to address interpretability challenges in practical applications. Zhu et al. (2023d) introduced a method to extract decision trees from deep reinforcement learning policies, transforming them into a tree-like structure that is easily comprehensible to humans and enhances model transparency. Gu (2024b) proposed π -Light, which constructs interpretable programmatic policies using domain-specific language and MCTS. SHAP (Shapley Additive Explanations) has been used to elucidate model decision-making processes (Jiang et al. 2022b) and analyze the influence of state features on action choices (Schreiber, 2022). Rizzo et al. (2019) trained a policy gradient model and used SHAP values to interpret how traffic flow features affect model decisions. Recently, Cai et al. (2025) proposed XLight, an interpretable MARL approach that addresses interpretability through three key aspects: careful design of RL components with provable equivalence between reward maximization and travel time minimization, implementation of regulatable control functions, and maximum entropy policy optimization. Their work demonstrates that interpretability can be achieved while maintaining competitive performance across various traffic scenarios. Despite these advances in improving the safety and interpretability of TSC systems, such as through analyzing feature importance to explain decision-making processes, existing research predominantly emphasizes interpretations at the individual feature level. A substantial gap persists in the analysis of collaborative strategies among multiple agents, resulting in a limited understanding of strategy interpretation in multi-agent traffic environments.

5.5. Partial sensor availability

In TSC, RL methods often rely on sensor data for state estimation and control decisions. However, in real-world applications, sensors may fail, experience communication interruptions, or have limited coverage, leading to data loss that affects TSC system performance. Partial observability and data loss can result in intelligent traffic signal controllers lacking complete and accurate

vehicle state information, leading to suboptimal decisions and decreased TSC efficiency. This issue is critical, as missing data can undermine coordinated control across multiple intersections, hindering optimal use of dynamic traffic flow characteristics (Xu, 2024). Furthermore, missing data introduce uncertainty, making the system more vulnerable when making signal control decisions, particularly in large-scale networks or unexpected traffic conditions (Shi, 2024).

To address the challenges of partial observability or data loss, various studies have explored different strategies, each with its own strengths and limitations. Xu (2024) proposed a state-space estimation method using WGAN to impute missing data within the state space, ensuring data integrity and improving multi-intersection TSC. This approach was shown to be particularly effective in maintaining robust control under diverse types of missing data. Boukerche et al. (2022) tackled data transmission delays with a new state and reward model, aiming to reduce discrepancies between real-time and delayed states. Their method enhanced the performance of collaborative signal control but showed limitations in environments with severe delays, where adjustments might still lag.

Mei (2023) proposed two approaches: imputing missing traffic states for adaptive control and filling in both states and rewards to train RL agents for robust performance under varying data loss rates. Fang (2023a) developed a dual-stream fused fuzzy deep neural network, combining fuzzy inference with deep learning to mitigate uncertainty and noise in partially observed environments, enhancing robustness and generalization in multi-agent learning. Zhang (2021) proposed models focusing on partial detection, introducing state representations and reward functions to adapt to low detection rates, showing significant improvements in reducing waiting times. However, these methods might not perform as well in scenarios with very low detection rates, where data sparsity becomes a critical issue. Ducrocq and Farhi (2023) designed discrete traffic state encoding (DTSE) and a reward function based on total squared delay to address low detection rates of connected vehicles in partially observable environments. Yang (2024) used RL to coordinate signal responses during malfunctions, with influence-aware state and reward modules allowing agents to adjust strategies collaboratively to reduce congestion. This highlighted how coordination can be designed for both optimal performance and resilience during disruptions.

5.6. Fairness

Fairness in TSC concerns ensuring that all road users receive equitable access to resources at traffic lights, preventing certain vehicles from experiencing excessive wait times. Du (2024) introduced a novel fairness metric that evaluates scheduling equity based on vehicle waiting times, penalizing phases with excessive delays. This targeted approach ensures that fairness is only enforced when waiting times exceed a threshold, preventing unnecessary performance sacrifices. The experimental results reveal that FELight effectively reduces prolonged waits for certain directions, achieving a fairer traffic distribution. Moreover, it maintains overall traffic efficiency, demonstrating its advantage over other fairness-oriented methods that often compromise system performance for equity. This balance makes FELight particularly suitable for real-world scenarios with diverse traffic conditions.

5.7. Transferability

The core of transferability in TSC lies in adapting a model from one scenario to another while maintaining robust performance

Table 5. Summary of RL-TSC methods addressing complex traffic scenarios (excluding mixed traffic flows) and practical challenges.

References	Scenario	Main advantages	Main limitations
Berbar (2022); Sun (2023); Gao et al. (2024); Li (2024a); Dai (2024b); Zhang (2025)	Integrated optimization of CAV and TSC systems	Simultaneous control of signal phases and vehicle speed; enhanced coordination between traffic signals and vehicle platoons; optimization of fuel consumption and travel efficiency	High computational complexity; extensive communication infrastructure requirements; limited applicability in mixed traffic with human-driven vehicles
Haydari et al. (2021); Muller and Sabatelli (2022); Ying (2022); Xu (2023a); Ren (2024b,c); Xu (2025)	System security	Prevention of hazardous decisions through action shielding; enhanced robustness against adversarial attacks and abnormal data; privacy protection for vehicle information	System flexibility limitations; computational overhead for security mechanisms; trade-offs between security measures and control performance
Rizzo et al. (2019); Schreiber (2022); Jiang et al. (2022b); Zhu et al. (2023d); Gu (2024b); Cai et al. (2025)	Interpretability	Enhanced model transparency and human-comprehensible decision logic; feature importance analysis for explaining decisions; maintained performance while achieving interpretability	Potential precision loss during interpretation; complex construction processes for interpretable models; limited interpretation of multi-agent collaborative strategies
Zhang (2021); Boukerche et al. (2022); Ducrocq and Farhi (2023); Mei (2023); Fang (2023a); Shi (2024); Xu (2024)	Partial sensor availability	Robust control under diverse missing data conditions; effective state estimation techniques; enhanced adaptability in environments with data loss	High computational requirements; performance degradation under extremely low detection rates; challenges in real-time implementation with complex imputation methods
Du (2024)	Fairness	Reduced directional waiting time disparities; enhanced scheduling equity; maintained overall traffic efficiency while improving fairness	Difficulty balancing fairness and efficiency objectives; additional complexity in reward design; potential performance trade-offs in highly congested scenarios
Liang (2022); Lu (2024); Jiang (2024a); Wang (2024a)	Transferability	Enhanced model generalization across different intersection structures; unified representation methods for diverse traffic networks; effective learning transfer between heterogeneous environments	Simplification of complex topological features; simulation-to-reality deployment challenges; high computational demands for some standardization approaches
Han (2023)	Action hysteresis	Reduced impact of control lag through traffic state prediction; enhanced safety with delayed control actions; maintained control effectiveness despite action constraints	Strong dependency on prediction accuracy; additional computational overhead for forecasting; potential responsiveness issues in rapidly changing traffic conditions

across different intersection structures. This requires constructing a universal state representation applicable across diverse intersection layouts.

Jiang (2024a) and Lu (2024) converted various topological features of intersections—such as entry and exit directions, lane states, and traffic signal information—into a unified feature vector and constructed a standardized intersection matrix. This approach maps different types of intersections into feature representations with consistent dimensions, ensuring that the model can handle diverse intersection structures and maintain a consistent input format across multiple scenarios, thereby enhancing the model’s generalizability and transferability. Similarly, Wang (2024a) employed an ‘intersection matrix’ approach to standardize the representation of different intersections, where fixed dimensions depict vehicle flow, lane counts, signal phases, and other relevant details. By standardizing varied intersection structures into a consistent matrix representation, the model preserves a uniform input format across heterogeneous environments. Additionally, Wang enhanced the model’s capability to adapt to different traffic conditions and unseen scenarios by leveraging traffic state augmentation techniques. These techniques include the use of intersection rotational symmetry and the relative positioning of vehicles, further strengthening the model’s generalizability and transferability. Meanwhile, Liang (2022) employed a ‘lane-cell’ model, which decomposes phase actions in TSC into lane-level option combinations, thereby allowing each lane cell to independently make decisions. This

approach captures specific state features of each lane while ensuring that different intersection topologies can be represented within a unified action space. However, methods that use detailed lane-level representations can be computationally demanding, potentially limiting real-time feasibility, especially in cities with limited infrastructure. While these models excel in simulation-based environments, bridging the gap to real-world deployment remains a challenge and requires further exploration.

5.8. Action hysteresis

In practical TSC, abrupt changes in traffic lights may lead to accidents, necessitating the incorporation of delays in control actions to enhance safety. However, these delays can compromise the effectiveness of control strategies. To address the issue of action delays, Han (2023) introduced a method that forecasts short-term future traffic states in real time. These predictions inform signal control decisions, helping to reduce the impact of control lags. The approach uses a policy-based model for TSC, optimizing actions based on these forecasts to maintain effective control when signals change. To ensure real-time responsiveness and efficiency, the method incorporates a local-view graph to collect traffic information from multiple sources and employs an advanced attention mechanism for feature extraction, enhancing the model’s responsiveness to traffic variations and minimizing efficiency losses due to delays.

While Table 4 summarizes the state and reward settings for mixed traffic flow studies, Table 5 provides a comprehensive

comparison of RL-TSC methods addressing other complex traffic scenarios (CAV-TSC integration) and practical challenges (system security, interpretability, partial sensor availability, fairness, transferability, and action hysteresis), highlighting their RL approaches, main advantages, and limitations.

6. Challenges and future directions

6.1. Scalability

Overview of the challenge

Scalability in RL-TSC remains a significant challenge within large-scale multimodal urban transportation networks. Recent research has employed GNNs to manage traffic flow across multiple intersections; however, most studies fail to fully account for the complexity of real-world multimodal traffic systems. Urban transportation networks encompass a wide range of participants, such as private vehicles, buses, pedestrians, cyclists, and freight trucks. Each type of participant exhibits unique behavioral traits that contribute to intricate and diverse traffic flow patterns. This complexity presents significant challenges in designing scalable models for coordinating large-scale networks, especially in managing multimodal traffic flows. The challenges associated with scalability can be categorized into several aspects:

- 1) Complexity of multimodal traffic flow: Numerous studies have explored multi-intersection coordination using GNNs, yet few have adequately addressed the challenges of large-scale multimodal traffic networks. Different types of traffic participants exhibit distinctive behaviors and interactions, resulting in complex traffic dynamics. This diversity complicates the ability of models to accurately capture these behaviors and their influence on traffic flow, thus hindering the optimization of TSC.
- 2) Difficulties in cross-network policy transfer and generalization: Transferring and generalizing policies across different traffic networks is challenging. While some studies have attempted to construct universal state representations, multimodal traffic increases the complexity of conditions, intersection layouts, and vehicle behaviors. These complexities not only challenge model performance but may also reduce coordination efficacy when applied to new networks. Particularly, when network structures undergo significant changes, maintaining effective coordination and policy generalization becomes increasingly difficult.
- 3) Designing suitable architectures and coordination strategies for large-scale road networks remains an open challenge. This challenge is intrinsically linked to and significantly exacerbated by the POMDP inherent in MARL-TSC, as discussed in Section 3.2.2. As the number of intersections (agents) increases, the portion of the overall network state hidden from any individual agent grows substantially. Each agent must make decisions based on increasingly localized and incomplete information, making decentralized coordination significantly more complex and fragile. Approaches like centralized training with decentralized execution (CTDE) face communication efficiency challenges in large road networks, which can be magnified by the need to potentially share richer information (e.g. belief states, more detailed local observations) to mitigate partial observability, demanding highly efficient communication protocols. Fully decentralized architectures offer promise in reducing communication load and enhancing scalability. However, they amplify the coordination challenge under partial observability, as

agents must autonomously coordinate based solely on local, incomplete information without central guidance or full network state visibility. Standard MARL approaches may struggle to learn effective global strategies due to this compounding effect of partial observability across many agents, risking either suboptimal decentralized behavior or paralysis if coordination fails.

Future directions

- 1) Future research could focus on creating more expressive heterogeneous graph neural networks or hypergraphs to capture the behaviors and interactions of diverse traffic participants, thereby enhancing the performance of RL models in large-scale multimodal traffic environments.
- 2) Further investigation into cross-network policy transfer methods is needed, particularly strategies that adapt to changes induced by multimodal traffic flow. Developing generalized policies that can accommodate varying traffic conditions and intersection layouts would improve the robustness and transferability of RL models across diverse urban environments.
- 3) Future research should focus on developing MARL methods that are both inherently scalable and robust to the severe partial observability encountered in large-scale TSC networks. This includes exploring advanced hierarchical MARL structures, possibly incorporating concepts like regionally adaptive partitioning, where higher-level units perform coarse-grained coordination based on aggregated or inferred information, while local agents handle fine control. Crucially, work is needed on scalable methods for decentralized state/belief estimation, efficient communication protocols designed for POMDP settings (e.g. conveying belief summaries or crucial neighbour information), and policy learning algorithms that explicitly leverage the POMDP structure (e.g. memory-based policies, planning under uncertainty) without succumbing to prohibitive complexity. Edge computing can serve as crucial infrastructure to support these scalable, decentralized approaches, enabling computation closer to intersections, reducing latency, and enhancing responsiveness. Creating architectures that effectively balance local reactivity with network-level coordination under significant partial observability remains a key objective for deploying MARL-TSC in large, real-world cities.

6.2. Real-time adaptation

Overview of the challenge

In practical TSC, traffic flow and environmental conditions are subject to real-time fluctuations, influenced by accidents, weather, road construction, and various unforeseen events. Hence, RL models must possess real-time adaptability, allowing them to swiftly respond to environmental changes and adjust signal control strategies accordingly. However, existing models' real-time adaptability encounters several challenges, primarily in the following areas:

- 1) Bottlenecks in large-scale real-time data processing and decision-making: The advancement of smart cities allows TSC systems to collect vast real-time data from various IoT devices, such as cameras, road sensors, and CAVs. This influx of data provides RL models with rich environmental information; however, swiftly extracting valuable insights,

making decisions, and issuing timely adjustments across large-scale networks remain significant challenges. Current models often struggle with limited computational resources, data latency, and noisy information when processing real-time data. These issues hinder the efficiency and accuracy of real-time decision-making.

- 2) Robustness in handling unknown scenarios: In real-world traffic, unforeseen scenarios frequently occur, such as large-scale events, natural disasters, and traffic accidents—conditions often absent in simulations. Therefore, models need the capacity to swiftly learn and adapt to new contexts. However, most current RL models, trained on existing datasets, struggle to handle entirely novel scenarios, often resulting in instability or suboptimal control during deployment.

Future directions

To enhance the real-time adaptability of RL models in practical TSC, future research can explore the following directions:

- 1) Given the influx of large-scale, multi-source real-time data, future research should develop methods to efficiently extract and utilize data, reducing processing and decision-making time. For instance, designing real-time dimensionality reduction and feature extraction algorithms can swiftly identify relevant information from large data streams, simplifying model inputs. Additionally, leveraging self-supervised learning and data compression can enhance real-time decision-making efficiency without compromising crucial information.
- 2) To enhance model adaptability to unforeseen situations, future research can explore techniques such as meta-learning and the integration of reinforcement learning with LLMs. Meta-learning enables rapid adaptation to new contexts by learning from previously unseen events and environments, while integrating LLMs' knowledge can improve the understanding of complex scenarios, thereby enhancing decision-making during emergencies. Additionally, employing scenario prediction and simulation techniques can pre-train models for extreme conditions, increasing their robustness against unknown situations.

6.3. Sim-to-real gap

Overview of the challenge

The application of RL-TSC relies on simulation environments for training and testing. However, the complexities of real-world traffic often exceed simulator capabilities, leading to a 'Sim-to-Real Gap,' where models perform well in simulations but struggle in actual deployment. This challenge is evident in the following aspects:

- 1) Limitations of simulators: Current traffic simulators fail to fully replicate real-world traffic dynamics, vehicle behaviors, and pedestrian interactions. For instance, traffic flow and vehicle behavior in simulations are based on preset rules, lacking the uncertainties of human driving and environmental factors in real traffic. Consequently, RL models learn from idealized scenarios and often fail to maintain stable performance in real-world conditions.
- 2) Challenges in policy transfer from simulation to reality: RL models require pre-training in simulations before fine-tuning in real-world scenarios. However, direct online learning in real traffic is costly and poses safety risks. Therefore,

the challenge is to fine-tune these models for real environments without disrupting traffic flow.

Future directions

- 1) Future research should develop high-fidelity traffic simulators to accurately model the diversity and uncertainties of real traffic. This includes refined traffic flow models that incorporate various participants (pedestrians, cyclists, buses, emergency vehicles) and complex driving behaviors, along with simulating varying weather conditions and road events. Such enhancements would make simulators more reflective of real-world scenarios, thereby bridging the Sim-to-Real Gap and enhancing model applicability to real environments. Efforts like SynTraC (Chen, 2024), which utilized simulators to generate realistic image-based data, represented a step in this direction, though further improvements in simulator fidelity and data diversity are needed.
- 2) Leveraging real-world traffic data and integrating it with simulated data is another effective approach to narrowing the Sim-to-Real Gap. Infusing real traffic flow, vehicle trajectories, and signal data into simulations can significantly improve the model's perception of real-world dynamics. Moreover, incorporating real-time IoT data (cameras, sensors, connected vehicles) can enrich training inputs and enhance model adaptability to diverse conditions.
- 3) A feasible strategy to mitigate the Sim-to-Real Gap is using pre-collected offline data for training, circumventing direct simulator training. Offline reinforcement learning allows the model to learn policies from static data, providing preliminary decision-making capabilities akin to pre-training. Subsequent fine-tuning through online learning in real traffic can adjust strategies, enhancing real-world adaptability. Techniques like domain adaptation or specific Sim2Real methods like Grounded Action Transformation proposed by Da (2023b) learning can optimize policies with limited online interactions, improving generalization while reducing deployment costs and safety risks.

6.4. Integration with other urban systems for coordinated optimization

Overview of the challenge

Urban traffic management transcends the isolated problem of TSC; it is an intricate endeavor intertwined with numerous urban subsystems. These include public transit scheduling, pedestrian and non-motorized traffic management, priority control for emergency vehicles, and the broader orchestration of urban resources and IoT devices. In recent years, the rise of innovative transportation modes has presented RL-based signal control systems with the imperative to achieve coupled optimization with these intelligent transportation systems. However, the intricate interrelationships among these subsystems pose a significant challenge in efficiently coordinating them to ensure that TSC strategies align with broader urban traffic optimization objectives.

- 1) Limitations and requirements of multi-objective optimization: While existing research has incorporated multi-objective optimization into RL to balance goals such as traffic flow, pedestrian safety, and public transport priority, most studies focus on a limited set of objectives or isolated systems. In reality, urban traffic involves far more complex and interdependent goals; for instance, ensuring the safety

of pedestrians and cyclists while simultaneously optimizing the punctuality of bus routes and the flow efficiency for private vehicles. The challenge lies in dynamically balancing these diverse and potentially conflicting objectives to develop a unified optimization strategy—a problem that remains inadequately addressed.

- 2) Modeling of collaborative and competitive: Within multi-agent TSC, there exist complex collaborative and competitive dynamics among various traffic participants, including traffic lights, ride-hailing systems, on-demand public transport, traditional buses, private vehicles, and pedestrians. Current research has begun to integrate these interactions into RL decision-making frameworks, yet there remains a lack of detailed modeling of these multi-participant relationships. For example, ride-hailing services and on-demand transit often compete for passenger choices, exacerbating traffic peaks on specific routes and times. Furthermore, signal control must provide appropriate prioritization for these diverse systems. In emergencies, priority needs to be given to emergency vehicles, while also considering the time-sensitive demands of ride-hailing and on-demand transit passengers. During peak hours, attention must be paid to both the punctuality of buses and the safety and efficiency of pedestrian crossings. Such dynamic collaborative and competitive interactions not only exist within individual systems but are also compounded in inter-system dynamics.

Future directions

- 1) Future research should advance dynamic multi-objective optimization methods to adapt to changing urban traffic demands. TSC objectives vary by time and context—prioritizing efficiency during rush hours and safety at night, for example. Therefore, developing methods for dynamic weight adjustment is crucial to balance and optimize various goals, achieving holistic traffic management.
- 2) To better coordinate interactions among diverse traffic participants, future research should explore sophisticated game models that capture multi-participant dynamics and objectives. Refining the modeling of collaborative and competitive relationships among signals, buses, private vehicles, pedestrians, and emergency vehicles can significantly enhance RL model performance in multi-system coordination. For example, employing theories like Nash equilibrium and mean-field games to model interactions between traffic signals and transportation systems can balance priority allocation, optimizing overall traffic flow.

6.5. Real-world deployment challenges

Overview of the challenge

While RL-based TSC shows promising results in simulations, its real-world deployment faces unique challenges that go beyond technical algorithm development. The ultimate goal of translating these methods into practical urban infrastructure requires addressing three critical barriers:

- 1) Data availability and quality constraints: Real-world deployment demands high-fidelity sensor networks capable of providing continuous multimodal data streams (vehicular trajectories, pedestrian flows, emergency vehicle detection, etc.). Many existing urban infrastructures lack the necessary

sensing density and data standardization required for RL model operation.

- 2) Safety-critical system integration: Traffic signals are safety-critical infrastructure where control failures can lead to catastrophic consequences. This imposes stringent requirements on RL systems' interpretability, fail-safe mechanisms, and compliance with existing traffic regulations—aspects often underexplored in academic research.
- 3) Institutional and ecosystem barriers: Successful deployment requires collaboration across transportation authorities, technology providers, and urban planners. Current institutional frameworks often lack protocols for continuous learning systems in critical infrastructure, raising challenges in liability assignment, update protocols, and performance auditing.

Future directions

- 1) Establishing public-private testbeds: Creating instrumented urban zones for controlled RL-TSC experimentation. These living laboratories should incorporate multimodal sensing, V2I communication, and mechanisms for controlled intervention.
- 2) Developing hybrid architecture frameworks: Combining RL's adaptive capabilities with traditional rule-based systems through safety layers and confidence-bound mechanisms. This could involve using RL for strategic phase selection while retaining deterministic timing constraints for minimum green times and pedestrian clearance intervals.
- 3) Advancing explainable RL for regulatory compliance: Creating visualization tools and verification protocols that map RL decisions to interpretable traffic parameters, enabling transportation authorities to audit and validate system behavior against established safety standards.

7. Available benchmarks

In recent years, research on RL-TSC has flourished, resulting in an increasing availability of open-source code. These resources have greatly facilitated model development and experimental validation. This chapter provides an overview of open-source libraries and publicly available datasets for TSC, offering valuable resources for research, as shown in Table 6. Many benchmarks include relevant datasets commonly used in academic research. Additionally, researchers targeting TSC optimization in specific regions can extract traffic networks from OpenStreetMap for simulation studies.

8. Conclusion

This paper provides a comprehensive review of the latest advancements in RL for TSC, emphasizing the performance of advanced RL methods and their potential applications in complex traffic scenarios. We conduct an in-depth analysis of both the engineering and applications of RL in TSC, including its integration with emerging technologies such as GNNs, offline reinforcement learning, and LLMs. Additionally, we review the extended research of RL-TSC, such as mixed traffic flow, real-world deployment, etc. We propose enhancements to the implementation of reinforcement learning in real-world traffic scenarios. Current research emphasizes the considerable potential of RL in TSC; however, fully harnessing this potential requires addressing persistent

Table 6. Open-source benchmarks and datasets for RL-TSC.

Name/Reference	Simulator	Link	Dataset description
RESCO (Ault and Sharon, 2021)	SUMO	https://github.com/Pi-Star-Lab/RESCO	TAPAS Cologne (Cologne city center) and InTAS (Ingolstadt, Germany).
LemgoRL (Müller, 2021)	SUMO	https://github.com/rl-ina/lemgorl	'OWL322' intersection in Lemgo, Germany.
Libsignal (Mei, 2024)	SUMO/CityFlow	https://github.com/DaRL-LibSignal/LibSignal	Jinan (3 x 4), Hangzhou (1 x 1, 4 x 4), Grid (4 x 4), Atlanda (1 x 3), and Manhattan (7 x 28).
CBlab (Liang, 2023)	CBEngine	https://github.com/caradryanl/CityBrainLab	Road networks from 100 major cities worldwide (sourced from OpenStreetMap).

challenges, such as scalability in large-scale networks, real-time adaptability, bridging the sim-to-real gap, and optimizing coordination across various traffic systems. Future research directions include enhancing RL model efficiency, integrating multi-source data, advancing cross-network transferability, improving practical deployment in traffic environments, and developing models that account for both collaborative and competitive interactions with public transportation systems.

This paper provides a comprehensive review of the latest advancements in RL for TSC, emphasizing the performance of advanced RL methods and their potential applications in complex traffic scenarios. While theoretical advancements have progressed rapidly, their translation into widespread real-world implementation remains a critical priority and challenge for the research community. We conduct an in-depth analysis of both the engineering and applications of RL in TSC, including its integration with emerging technologies such as GNNs, offline reinforcement learning, and LLMs. Additionally, we review the extended research of RL-TSC, such as mixed traffic flow, real-world deployment challenges, etc. Current research emphasizes the considerable potential of RL in TSC; however, fully harnessing this potential requires addressing persistent challenges, such as scalability in large-scale networks, real-time adaptability, bridging the sim-to-real gap, and the development of safety-certified deployment frameworks, and most crucially, optimizing coordination across various traffic systems. Future research directions include enhancing RL model efficiency, transitioning from theoretical validation to field-tested solutions through integrating multi-source data, advancing cross-network transferability, establishing standardized real-world evaluation protocols, improving practical deployment in traffic environments, developing hybrid architectures that combine learning adaptability with regulatory compliance, and fostering institutional partnerships to enable scaled implementation.

Author contributions

Feng Xiao (Writing—Review & Editing, Conceptualization, Supervision, Funding acquisition), Jiaming Lu (Writing— Original Draft, Methodology, Formal analysis, Investigation, Visualization.), Lu Li (Writing—Review & Editing, Supervision), Wenwen Tu (Writing— Review & Editing, Validation), and Chaojing Li (Writing—Original Draft, Data Curation, Investigation)

Conflict of interest

Feng Xiao holds the position of Associate Editor for *Intelligent Transportation Infrastructure* and has not peer reviewed or made any editorial decisions for this paper.

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Appendix A. Abbreviations

As shown in Table A1.

Table A1. Glossary of key terms and abbreviations.

Term	Abbreviation
Reinforcement learning	RL
Artificial intelligence	AI
Traffic signal control	TSC
Multi-agent reinforcement learning	MARL
Reinforcement learning for traffic signal control	RL-TSC
Fixed-time signal control	FTSC
Traffic actuated signal control	TASC
Adaptive traffic signal control	ATSC
Markov decision process	MDP
Centralized training and decentralized execution	CTDE
Decentralized training and decentralized execution	DTDE
Deep Q-network	DQN
Proximal policy optimization	PPO
Soft actor-critic	SAC
Multi-agent proximal policy optimization	MAPPO
Q-mixing network	QMIX
Graph neural networks	GNNs
Spatio-temporal graph neural networks	ST-GNNs
Curriculum learning	CL
Offline reinforcement learning	Offline RL
Large language models	LLMs
Multi-objective reinforcement learning	MORL
Connected vehicles	CVs
Connected and autonomous vehicles	CAV
Human vehicles	HVs
Single-occupancy vehicles	SOVs
High-occupancy vehicles	HOVs
Vehicle-to-vehicle	V2V
Vehicle-to-infrastructure	V2I
Emergency vehicle	EMV
Transit Signal Priority	TSP
Generative adversarial networks	GANs
Wasserstein GANs	WGANs
Monte Carlo Tree Search	MCTS
Shapley additive explanations	SHAP
Partially detectable traffic signal control	PD-ITSC

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